

If the **WORKBOOK** menu item in the **VIEW** menu is activated, then all opened pages are displayed in Excel style (tabbed page list) at the bottom of the workspace. You can then use the mouse to switch back and forth between these pages – or the **CTRL + TAB** keyboard shortcut to sequence through all the pages. With three pages, the sequence starts at page 1 followed by page 2, page 3, back to page 1, etc.

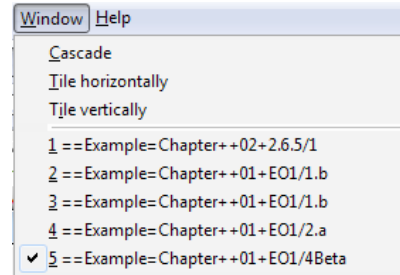


Fig. 4.18
Window menu

If several pages are open, you can use the **WIN-DOW** menu to go directly to the desired page.

To close open pages, simply select the pages in the page navigator and in the **PAGE** menu, select **CLOSE**. EPLAN then closes all selected pages.

4.1.2.3 Copy pages

When you create new pages, they are created without any page content (devices, graphics, etc.). To create a new page that uses the content of an existing page, you need to copy one or more pages. There are several ways to do this in EPLAN. You can use the standard Windows functions like **CTRL + C** (copy page) and **CTRL + V** (paste page), or the **PAGE** menu (or the page navigator popup menu).

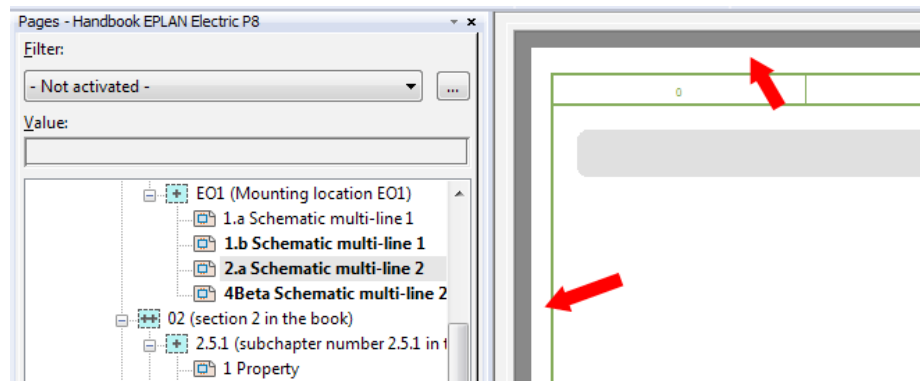
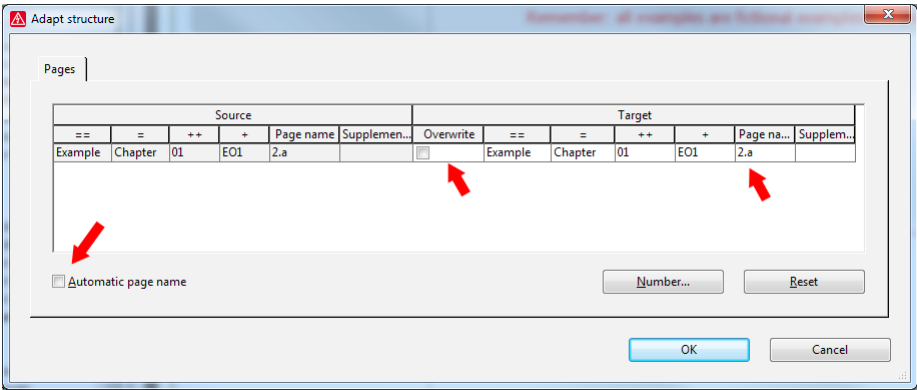


Fig. 4.19 Using **CTRL + M** to select a page

Use **CTRL + M** to select a page (outside the page navigator) in the graphical editor in order to copy it. EPLAN then draws a thick gray border around the page (to show it is selected).

When you press **CTRL + C** (standard Windows function for copying), EPLAN copies the page to the clipboard. Pressing **CTRL + V** (paste) retrieves the copied page from the clipboard and pastes it to a different location. EPLAN then opens the **ADAPT STRUCTURE** dialog.



Copying pages within a project and across projects

Fig. 4.20 Adapt structure dialog (without determining page name automatically)

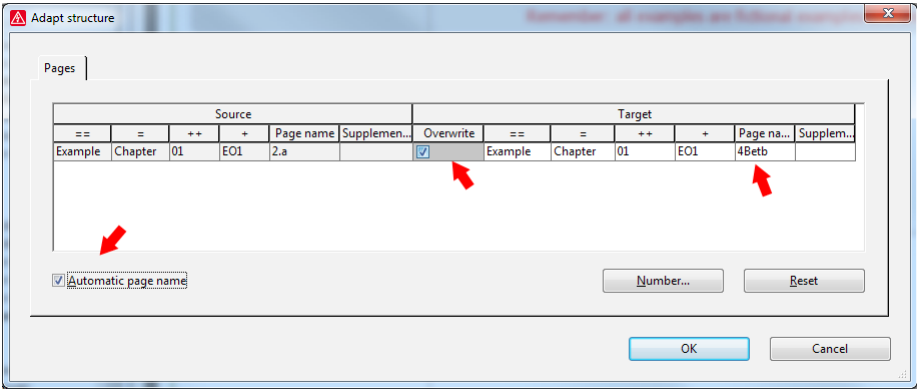


Fig. 4.21 Adapt structure dialog (with determining page name automatically)

Relevant entries, such as the assignment of identifiers or a new page name, can now be edited in this dialog. However, EPLAN can also determine the new page name automatically (*Automatic page name* setting). EPLAN always suggests the highest free page name available in the identifier structure selected. Once all entries are correct, you click OK to apply the page.

Depending on the settings and on whether there are any devices on the page to be copied), the INSERTION MODE dialog opens to allow you to number existing devices on the

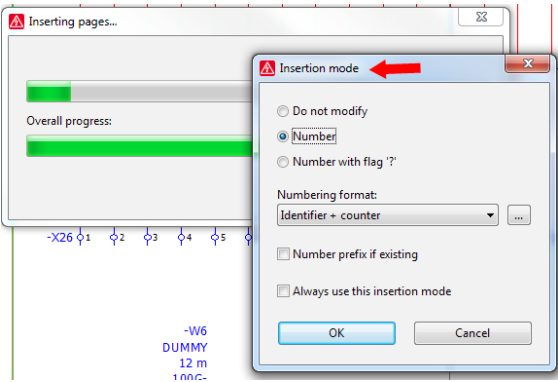


Fig. 4.22 Selecting the insertion mode

copied page. After selecting the desired setting, you exit the dialog by clicking OK. EPLAN then copies the (source) page, generates a new (target) page, numbers the devices according to the setting, and sorts the page into the page structure.

At this point, the ADAPT STRUCTURE dialog also lets you overwrite existing pages (using the **Target – Overwrite** column).

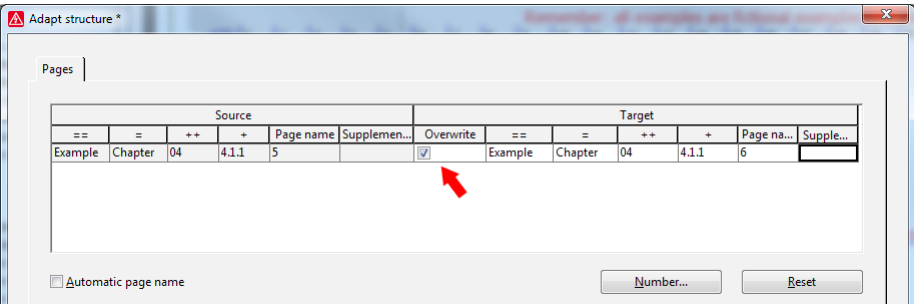


Fig. 4.23
Selecting Overwrite
option

Before actually overwriting pages EPLAN displays a prompt, which must be confirmed by clicking YES (copy action continues) or NO (copy action is canceled and EPLAN opens the page navigator once more).

Again the following applies when copying pages: Without asking for confirmation, EPLAN automatically sorts new identifiers into the existing structure (alphabetically or at the end, depending on the setting). They can be resorted later in structure identifier management.

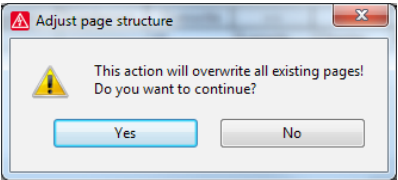


Fig. 4.24
Note regarding the
Overwrite action

Copying pages in the
page navigator

You can, of course, copy more than one page. In the page navigator, you can select multiple pages and use CTRL + C to copy them to the clipboard.

When you press CTRL + V (paste), EPLAN again opens the already-familiar ADAPT STRUCTURE dialog. After you make all desired entries and confirm any subsequent dialogs, these pages are inserted into the EPLAN project in the selected structure.

Page navigator List view

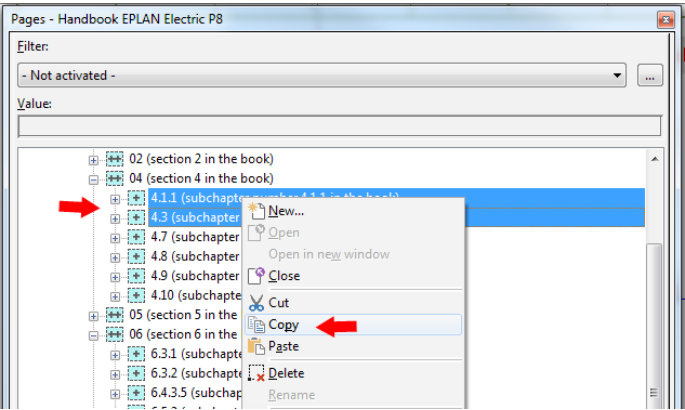


Fig. 4.25
Copying in the tree
view

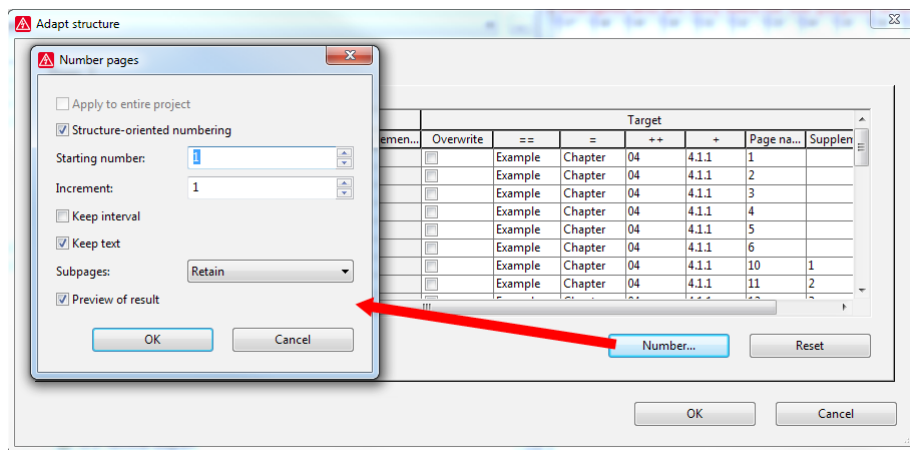


Fig. 4.26 Adapt structure dialog with Number pages option

Instead of **CTRL + C** and **CTRL + V**, you can also use the **COPY** and **PASTE** commands in the **page navigator** popup menu to copy pages. These two commands always relate to the selected pages. You can select and copy/paste pages from either the tree view or the list view.



TIP: **CTRL + M (SELECT PAGE)** is an important keyboard shortcut to remember. All page-related commands, i.e. copying a page, deleting a page or editing page properties, are implemented by pressing **CTRL + M** and then the subsequent action command.

In addition to copying pages within a project, you can also copy the pages of a project to the clipboard, switch to another project (which should be open in the page navigator) and there, paste the pages from the clipboard.

4.1.2.3.1 Number button

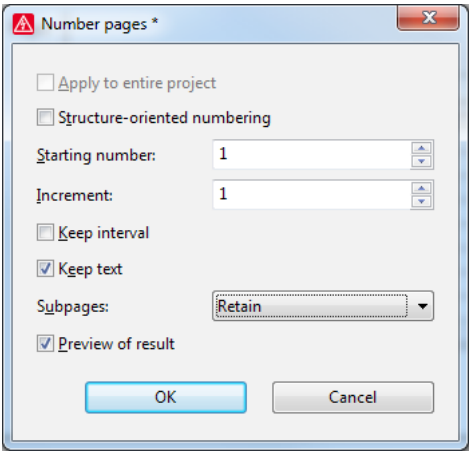
The option to re-number pages immediately when inserted offers some advantages. Page numbers do not have to be adjusted manually. Instead, if so desired, EPLAN can number the page numbers in an ascending order, for example, sorted by structure identifiers.



TIP: The preview of results should always be selected as well. This way, you can see the numbering results right away and, if necessary, reverse them via Cancel/Restore, without the changes being written to the project.

Two small examples will illustrate this. For setting 1, the start number 1 is selected, and no structure-based numbering. Text and subpages are retained. After you click the NUMBER PAGES dialog, EPLAN numbers all pages (across identifier structures).

Fig. 4.27
Number pages dialog,
setting 1

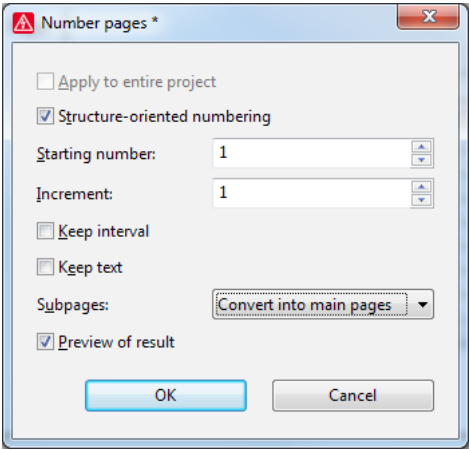


Target						
Overwrite	==	=	++	+	Page na...	Supplen
☐	Example	Chapter	04	4.1.1	5	
☐	Example	Chapter	04	4.1.1	6	
☒	Example	Chapter	04	4.1.1	7	1
☒	Example	Chapter	04	4.1.1	8	2
☒	Example	Chapter	04	4.1.1	9	3
☐	Example	Chapter	04	4.1.1	10	4
☐	Example	Chapter	04	4.1.1	11	5
☐	Example	Chapter	04	4.1.1	12	6

Fig. 4.28
Numbering result,
setting 1

A second example illustrates setting 2. Here, the settings are structure-oriented numbering, start number 1, do not retain text, and the subpages are to be converted to main pages. After you click the OK button, EPLAN breaks up the subpages, removes the alphabetical elements of a page number, and numbers the pages in an ascending order by structure identifiers, starting at 1.

Fig. 4.29
Number pages dialog,
setting 2



Target						
Overwrite	==	=	++	+	Page na...	Supplen
☐	Example	Chapter	04	4.1.1	14	8
☐	Example	Chapter	04	4.1.1	15	9
☐	Example	Chapter	04	4.1.1	16	10
☐	Example	Chapter	04	4.1.1	17	11
☐	Example	Chapter	04	4.1.1	18	12
☐	Example	Chapter	04	4.1.1	19	13
☐	Example	Chapter	04	4.1.1	20	14
☐	Example	Chapter	04	4.1.1	21	15

Fig. 4.30
Numbering result,
setting 2

4.1.2.3.2 Restore button

The RESTORE button restores the numbering to the values prior to the numbering.

Target						
Overwrite	==	=	++	+	Page na...	Supplem...
☐	Example	Chapter	04	4.1.1	14	8
☐	Example	Chapter	04	4.1.1	15	9
☐	Example	Chapter	04	4.1.1	16	10
☐	Example	Chapter	04	4.1.1	17	11
☐	Example	Chapter	04	4.1.1	18	12
☐	Example	Chapter	04	4.1.1	19	13
☐	Example	Chapter	04	4.1.1	20	14
☐	Example	Chapter	04	4.1.1	21	15
☐	Example	Chapter	04	4.1.1	22	16
☐	Example	Chapter	04	4.1.1	23	
☐	Example	Chapter	04	4.1.1	24	
☐	Example	Chapter	04	4.1.1	25	
☐	Example	Chapter	04	4.1.1	26	
☐	Example	Chapter	04	4.1.1	27	

Number...

Reset

Fig. 4.31
Page numbering
prior to restore

Target						
Overwrite	==	=	++	+	Page na...	Supplem...
☐	Example	Chapter	04	4.1.1	17	8
☐	Example	Chapter	04	4.1.1	18	9
☐	Example	Chapter	04	4.1.1	18.a	10
☐	Example	Chapter	04	4.1.1	19	11
☐	Example	Chapter	04	4.1.1	19.a	12
☐	Example	Chapter	04	4.1.1	20	13
☐	Example	Chapter	04	4.1.1	21	14
☐	Example	Chapter	04	4.1.1	22	15
☐	Example	Chapter	04	4.1.1	23	16
☐	Example	Chapter	04	4.3	1	
☐	Example	Chapter	04	4.3	2	
☐	Example	Chapter	04	4.3	3	
☐	Example	Chapter	04	4.3	4	
☐	Example	Chapter	04	4.3	5	

Number...

Reset

Fig. 4.32
Page numbering after
restore

4.1.2.4 Copy pages from/to

The counterpart to simple page copying (within a project and to other projects) is the **COPY FROM/TO...** function. Since you can open and edit several projects at once in the EPLAN page navigator, it is very useful to have a function that lets you conveniently copy pages from one project to another.

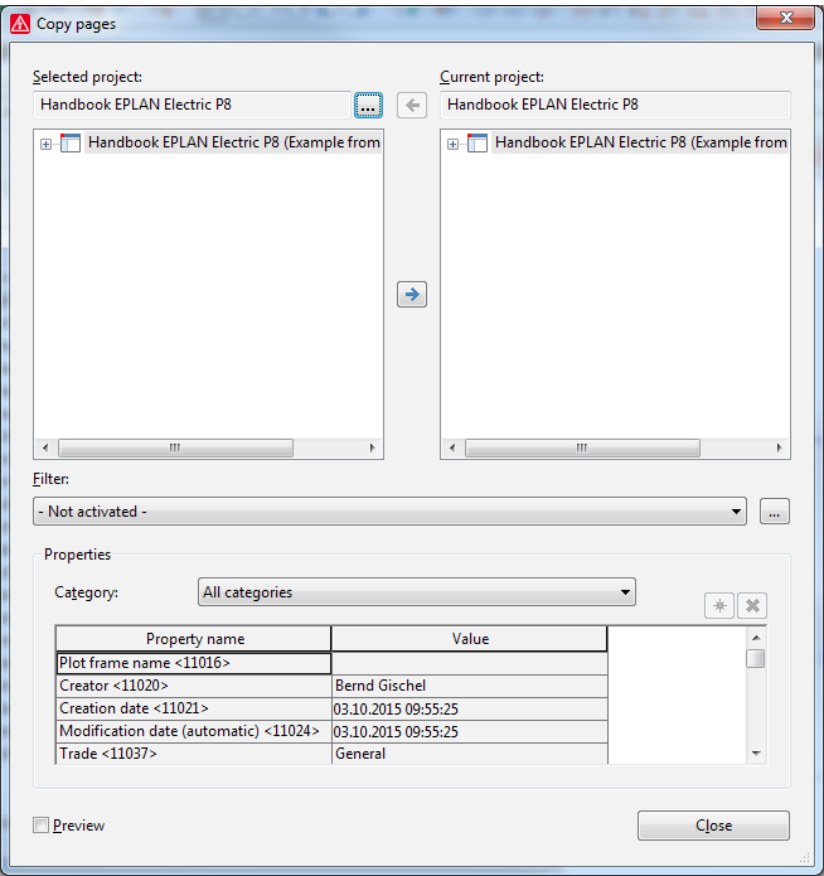


Fig. 4.33 Copy pages dialog

Copy pages from/to
function

The **COPY PAGE/PAGES FROM/TO** menu lets you copy pages from different projects (even from projects that are not open in the page navigator) to the current project. When you select **COPY PAGES FROM/TO** from the menu, EPLAN opens the **COPY PAGES** dialog. As already mentioned, a project does not need to be open in the page navigator in order to copy pages from it.

The lower area of the **COPY PAGES** dialog shows additional information about the selected page. A preview for the selected pages can also be activated. The right field in the **COPY PAGES** dialog is fixed and always indicates the *current project*. It cannot be changed. The

selected project in the left field is variable. You can use the  button to select and open any other project.

When you click , EPLAN opens the **PROJECT SELECTION** dialog. Initially, only projects currently open in the page navigator are listed.

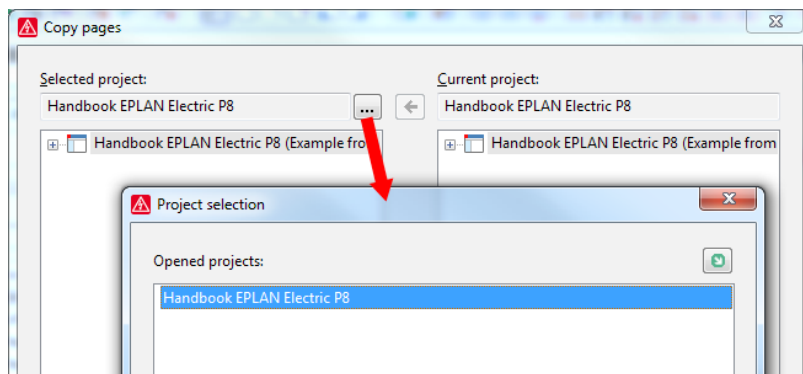


Fig. 4.34 Project selection dialog

When you click the  button (OPEN), EPLAN opens the **OPEN PROJECT** dialog. In this dialog, you can select and open one or more projects to be used in the **PROJECT SELECTION** dialog. First you select the project(s), then click **OPEN** to add the selected project(s) to the **PROJECT SELECTION** dialog.

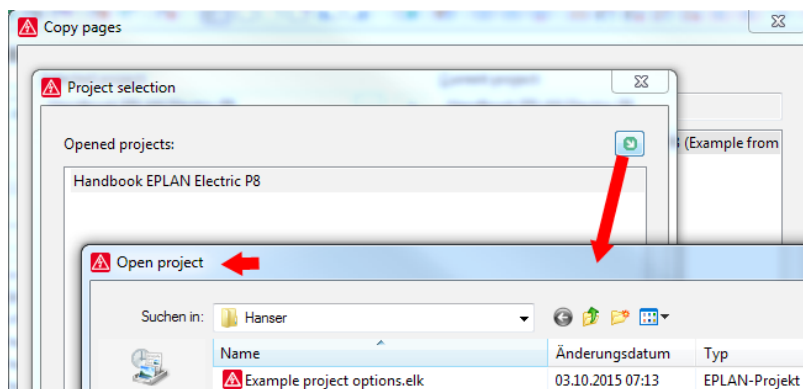


Fig. 4.35 Open project dialog

When you have finished, the selected projects are listed in the **PROJECT SELECTION** dialog ready for page copying and are also opened simultaneously in the page navigator.

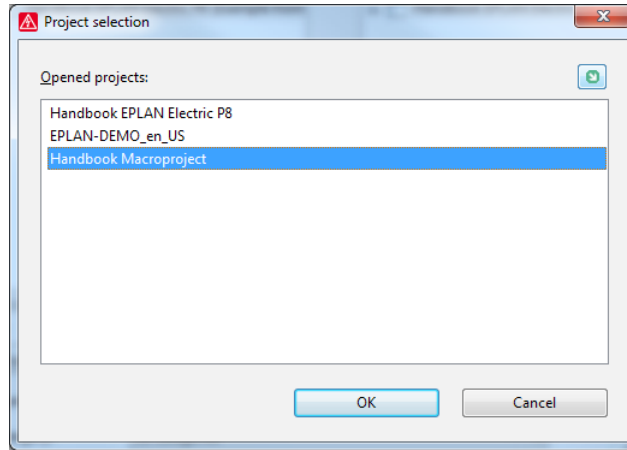


Fig. 4.36 Opened projects

You can now select the desired project in the **PROJECT SELECTION** dialog and add them to the **COPY PAGES** dialog by clicking the **OK** button.

NOTE: Only one project at a time can be added to the selection for the **COPY PAGES** dialog. However, EPLAN remembers the projects that were added to the **PROJECT SELECTION** dialog so that later copying can take place without having to re-select a project.

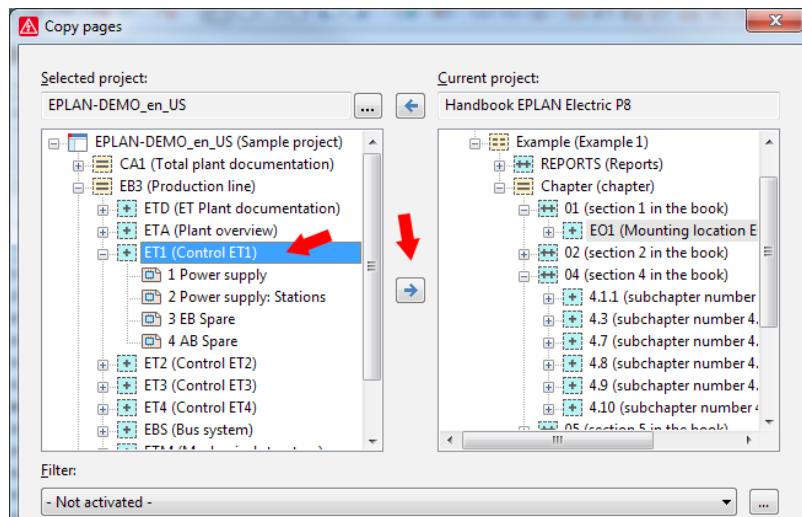


Fig. 4.37 Copying pages from a project

In the **COPY PAGES** dialog, on the left you *select* the pages you want to copy from the external project and use the button in the middle to copy (move) them to the current project. EPLAN then opens the **ADAPT STRUCTURE** dialog.

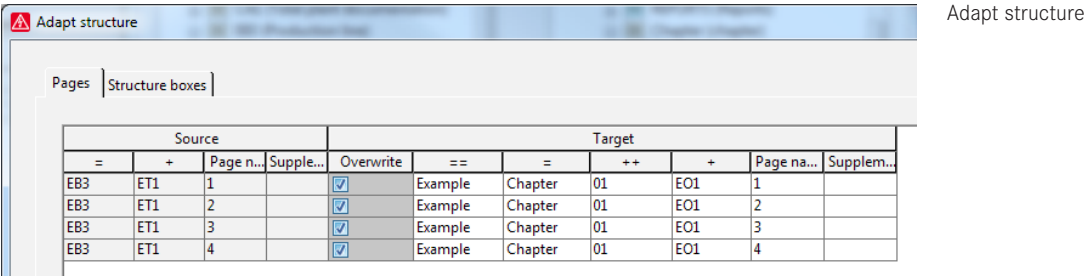


Fig. 4.38 Inserting pages into the page structure

In the **ADAPT STRUCTURE** dialog, you can adjust the page structure as desired.

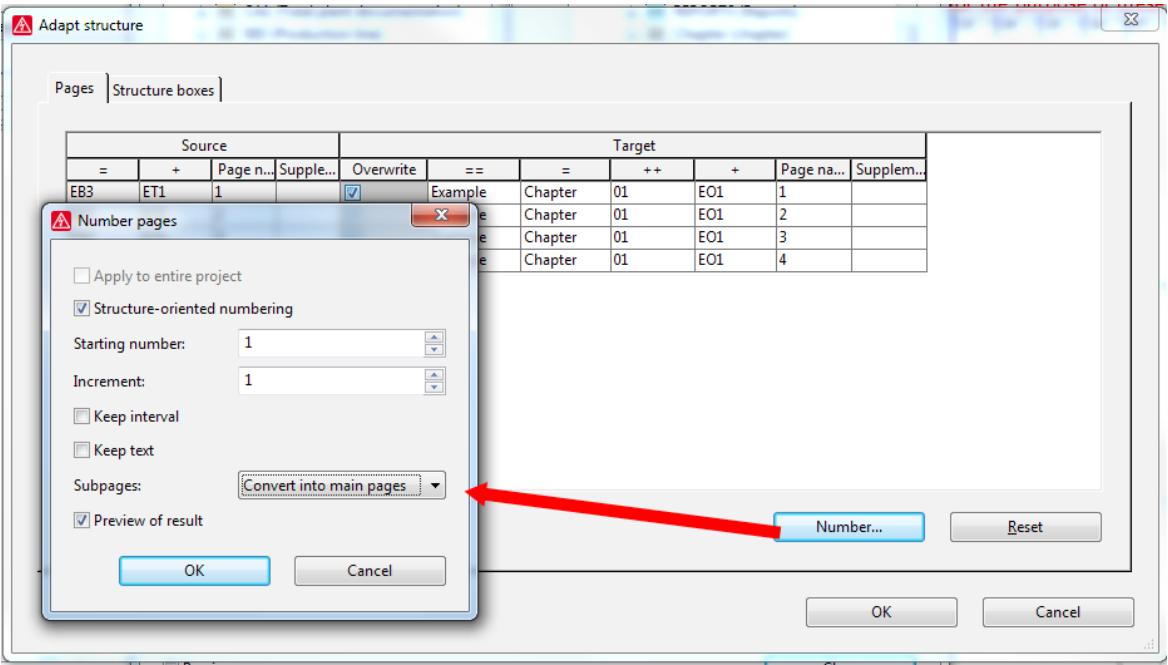


Fig. 4.39 Number pages

In the **NUMBER PAGES** dialog, it is possible to re-number the new pages according to specific settings. When you need to copy many subpages and convert them to main pages, for example.

After you click the **OK** button in the **ADAPT STRUCTURE** dialog, EPLAN copies the pages into the current project. Any follow-up dialogs, such as the **INSERTION MODE** dialog, must be confirmed as required.

This completes the copying of pages from external projects. EPLAN closes all dialogs and again focuses on the **COPY PAGES** dialog. You can now copy pages from other projects. If this is not required, you can exit the **COPY PAGES** dialog with the **CLOSE** button. EPLAN closes the dialog and all temporarily opened projects and returns to graphical editing of the currently opened project.

4.1.2.5 Rename pages

Pages in EPLAN are given a unique page name (page number). This page name **can** include alphanumeric characters. A page could have the name =ANL2+ORT2/14_Overview, for example, whereby the word “Overview” is part of the page name and not the page description.

The page name (number) is renamed using the **RENAME PAGE** menu item. In the **page navigator**, you can also simply select the page and press the **F2** key.

EPLAN then lets you change the page name (number) without directly opening the page properties of this page. Pressing **ENTER** saves the changed page name (number).

Renaming the page
name (number)

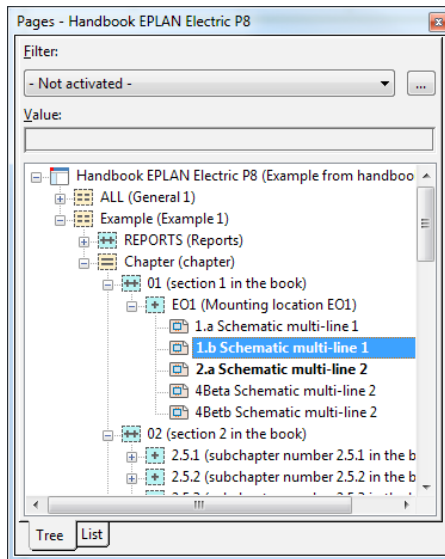


Fig. 4.40 Selecting a page

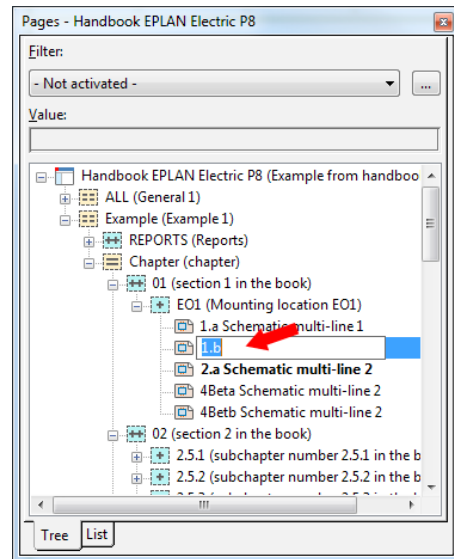


Fig. 4.41 Direct editing via F2 (page name with subpage)

4.1.2.6 Delete pages

But you don't just create or copy pages. Sometimes you need to delete them. In the popup menu of the page navigator, EPLAN provides the **DELETE** function for this purpose.

You are already familiar with the procedure. The pages to be deleted are selected in the page navigator and then deleted using the *Delete* function in the page navigator popup menu or by pressing the **DEL** key.

Before the actual deletion, EPLAN displays a **DELETE PAGES** dialog with a warning message asking if the page(s) should actually be deleted.



NOTE: I recommend that you read the messages carefully before **blindly** clicking YES or YES TO ALL.



NOTE: When deleting pages, it is important to know that there is a difference in how pages are selected in the page navigator.

In Fig. 4.42, the four pages have been selected to be deleted. Only these four pages will be deleted in this case. The warning message in the **DELETE PAGES** dialog is very clear: “Do you want to delete Page ... out of 4 selected pages?”

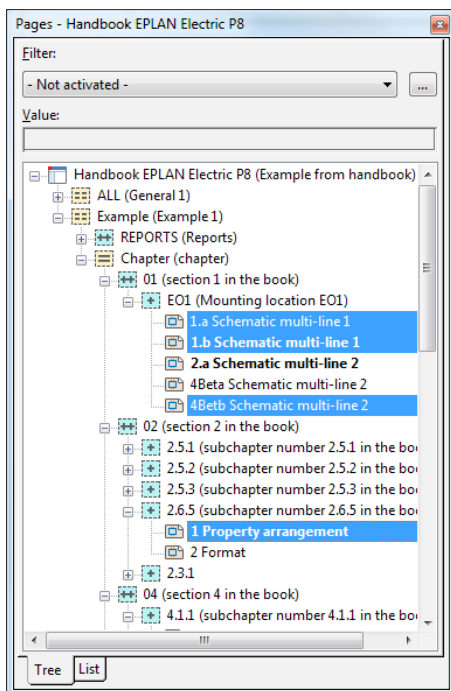


Fig. 4.42 Selection 1: 4 pages

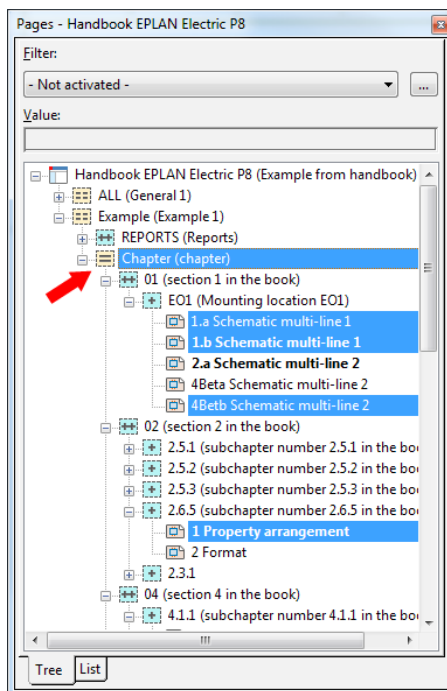


Fig. 4.43 Selection 2: Entire higher-level function (chapter)

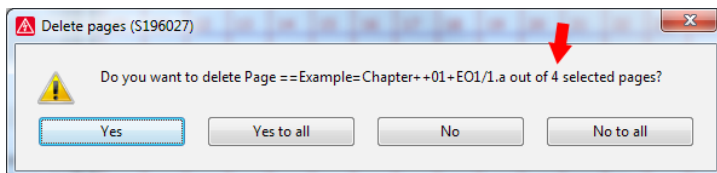


Fig. 4.44 Warning message: Delete 4 pages

In Fig. 4.43, however, the selection has been extended by mistake to the entire higher-level function =*Chapter*, because the higher-level function was selected in the **page navigator**.

If you try to delete these selected pages, EPLAN displays the familiar **DELETE PAGES** dialog with an appropriate warning message, but with the minor difference that the *number of pages to be deleted* is no longer correct.

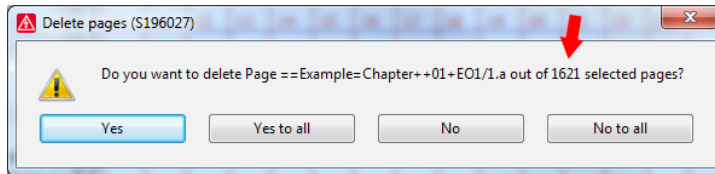


Fig. 4.45 Warning message: Delete 1621 pages? (instead of the desired 4)

If you unthinkingly confirm this dialog with **YES**, then EPLAN will delete the entire higher-level function = *Chapter* from the project. Unintentional deletion of pages is usually noticed immediately, since the page navigator is updated after a deletion and the missing pages, higher-level functions, or locations are (or should be) immediately obvious when editing a project. The usual UNDO command **CTRL + Z**, or the **UNDO** or **UNDO LIST** functions in the **EDIT** menu can then be used to restore the pages *immediately after deletion*.



WARNING: If the project was closed in the interim, the deleted pages are irretrievably lost!



NOTE: If the project name is selected, then EPLAN disables the **DELETE** function. This prevents accidental deletion of entire projects from the page navigator popup menu.

4.1.2.7 Close pages

This function is self-explanatory. It closes open pages. The **CLOSE** function is accessible in the **PAGE** menu and also via the page navigator popup menu.

It is possible to select several open pages in the page navigator individually and close them all at once. The **CLOSE** function only closes the project page(s) and not the project, not even if it is selected.



TIP: To close several or all opened pages in one go, simply select the project name and call the **CLOSE** command via the popup menu. This will close all opened page with a single click.

4.1.2.8 Edit page properties

To edit the page entries, the so-called *Page properties*, you need to select the relevant page in the page navigator and use the popup menu to call up the **PROPERTIES** function.

Of course, this can also be done without the page navigator, directly on the opened page using the **CTRL + M** and **CTRL + D** shortcut keys. **CTRL + M** selects again the page and then **CTRL + D** calls the **EDIT PROPERTIES** function.

EPLAN then opens the **PAGE PROPERTIES** dialog.

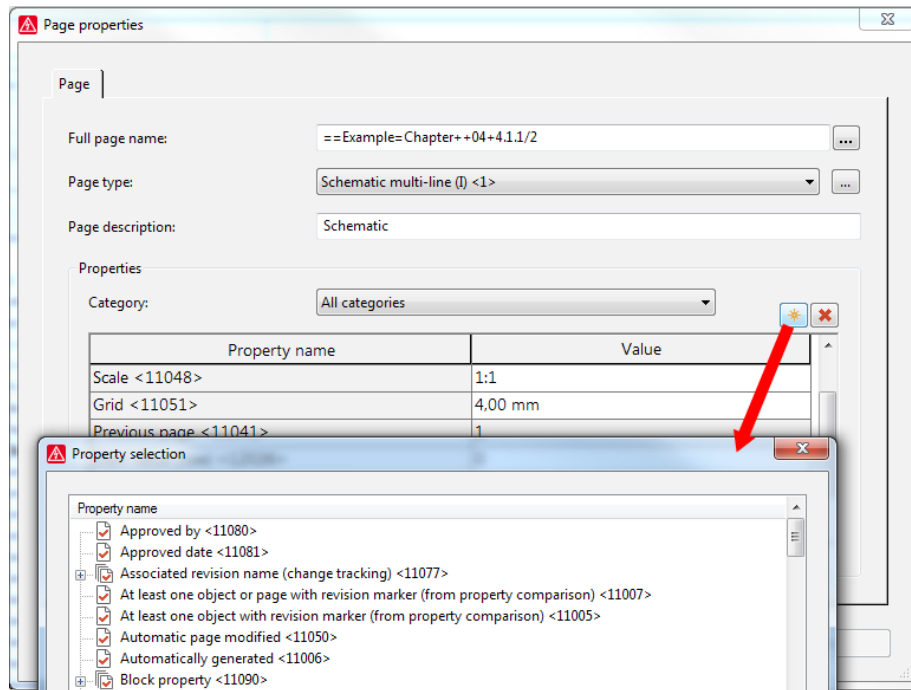


Fig. 4.46 Edit page properties dialog

You can now edit all the page properties in the **PAGE PROPERTIES** dialog. The *Full page name* can be changed (for example, you could change the page number from 2 to 3, which will move the page) or modify the *Page description*.

Edit page properties

The *Properties* area displays additional properties, such as page scale (*Scale <11048>*) or the currently set grid (*Grid <11051>*). These properties can all be edited when they are not grayed out.

New properties are added via the  button. When this button is clicked, EPLAN opens the **PROPERTY SELECTION** dialog. Here you can select the desired properties (multiple selections are possible) and apply them in the **Page properties** display by clicking the **OK** button.

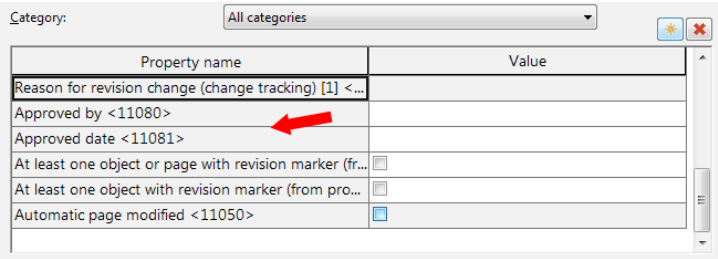


Fig. 4.47 Newly added properties

In the **page navigator**, you can edit more than one page at once, which is not the case for the **CTRL + M** and subsequent **CTRL + D** keyboard shortcuts.

If several pages are selected in the page navigator and the popup menu is then used to call the **PROPERTIES** function, then the properties of all selected pages can be edited in a single step.

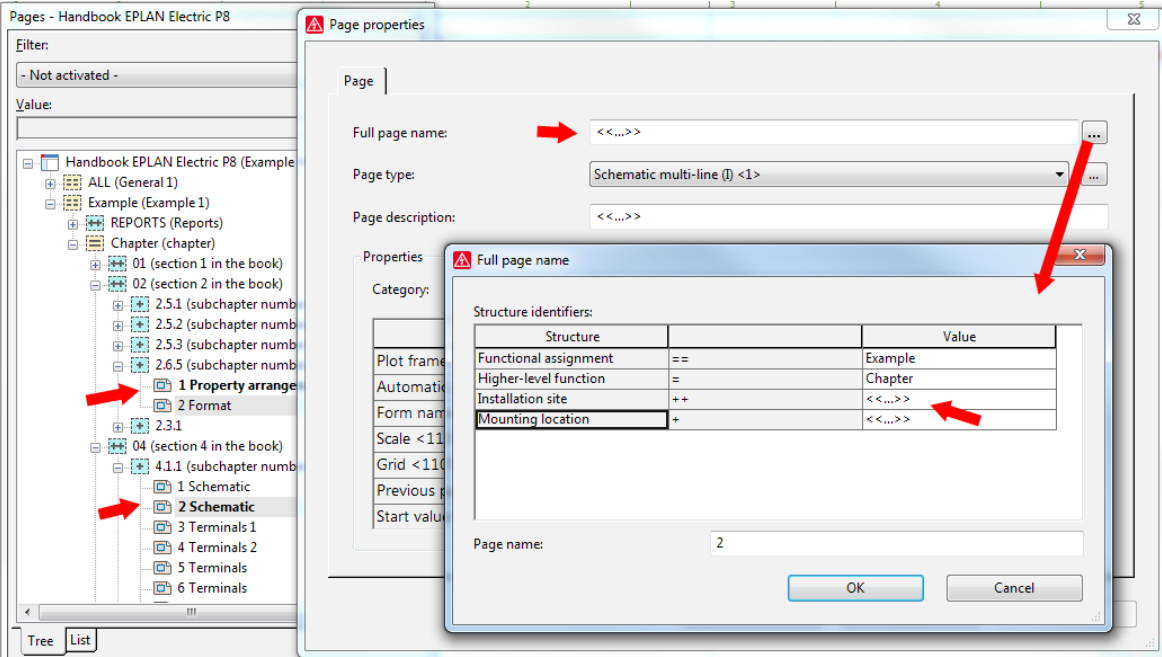


Fig. 4.48 Editing the properties of multiple pages

All properties that are the same in the selected pages are displayed in “plain text”. Examples would be the identifier (structure), scale or grid of the pages. If the page properties are different from each other, then they are displayed visually using the <<...>> string in the corresponding input fields.



NOTE: Care should be taken when changing fields containing the <<...>> string because when an entry in a field is changed, all other entries that are not the same on other pages are overwritten (without a prompt) with the new (changed) value.

When all entries are complete, you can exit the **PAGE PROPERTIES** dialog by clicking **OK**. EPLAN saves all entries and then closes the dialog.

4.1.2.9 Edit page properties directly

In addition to starting the **PAGE PROPERTIES** dialog via the page navigator popup menu or the **CTRL + M** and **CTRL + D** shortcut keys in order to adjust or modify properties such as the page description, etc., it is now possible to simply double-click an existing object (page description, customer name, graphical element) in the plot frame to have EPLAN open the page properties.

			Date	01.10.2015	Handbook EPLAN Electric P8 Example from handbook	
			Ed. by	Bernd Gischel		
			Appr	Bernd Gischel		
Modification	Date	Name	Original		Replacement for	

Fig. 4.49 Double-click any object in the plot frame to open page properties

But it has to be an area that contains content. Double-clicking an empty area will *not* open the page properties.

			Date	01.10.2015	Handbook EPLAN Electric P8 Example from handbook	
			Ed. by	Bernd Gischel		
			Appr	Bernd Gischel		
Modification	Date	Name	Original		Replacement for	Repla

Fig. 4.50 Empty areas do not work

4.1.3 Page navigator filter

In addition to the functions previously described, it is now also possible to directly activate a filter for previously created schemes and use the quick input functions in the page navigator and in other navigators.

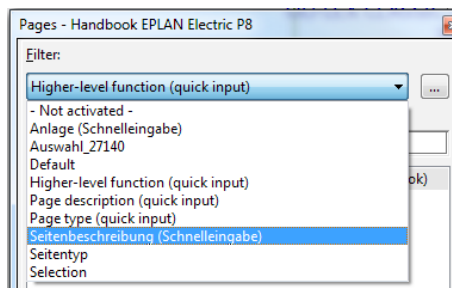


Fig. 4.51
Filter options

This lets you quickly and easily filter, for example, the page display by a specific **page type** (“quick entry”). To do so, you first create a filter scheme and then activate the **quick entry** column.

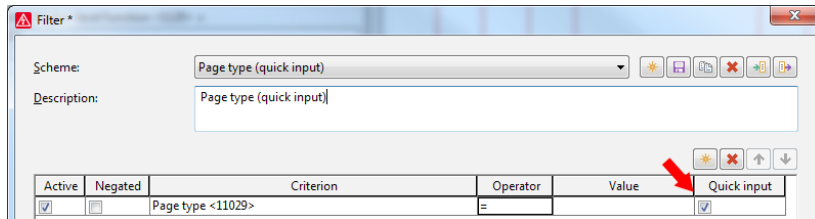


Fig. 4.52 Filtering by quick entry

When the filter has been activated, you can use the **Value selection: [criterion]** to execute a quick entry (**page type** in this example).

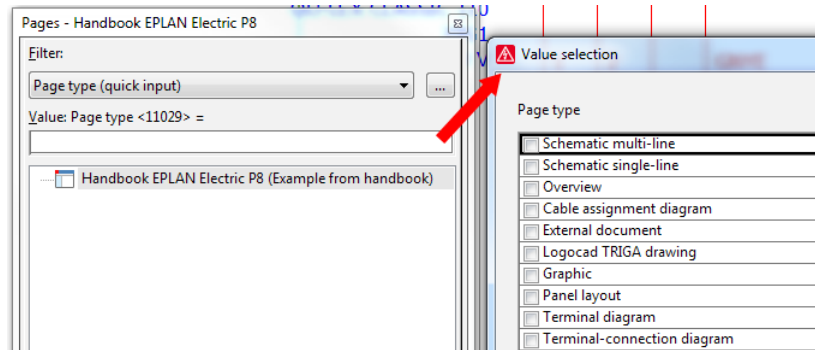


Fig. 4.53 Active page type filter

The main advantage of being able to combine filters with quick entry is that you do not have to create an extra filter scheme for each possibility. You only have to set one, with one criterion, like here in the **Page type**. You simply activate the required value, here the to-be-filtered **page type**.

EPLAN also offers further filter options. The detailed selection. It is not limited to filtering page properties, but you can also activate further filters, for example, for specific identifiers and similar elements in the page contents.

Apart from the functions described here, there are a number of other options in the page navigator. Some of these will be explained briefly in the following sections.

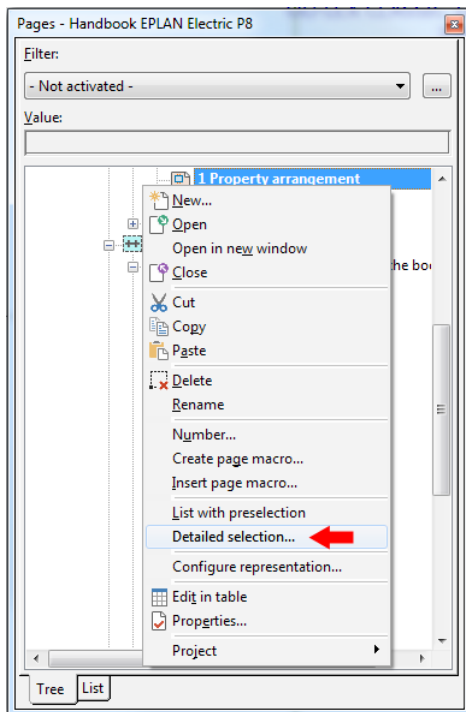


Fig. 4.54 Detailed selection

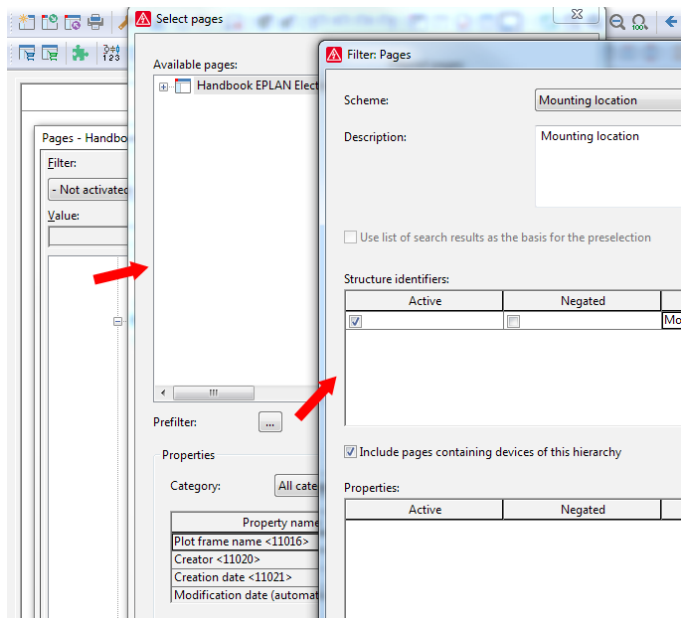


Fig. 4.55 Filter options

4.1.4 Edit in table

The edit in table function allows the user to edit/revise page property data quickly and conveniently, similar to the way it is done in Excel. To use this function, select the pages to be edited in the page navigator, and then call the **EDIT IN TABLE** function via the popup menu.

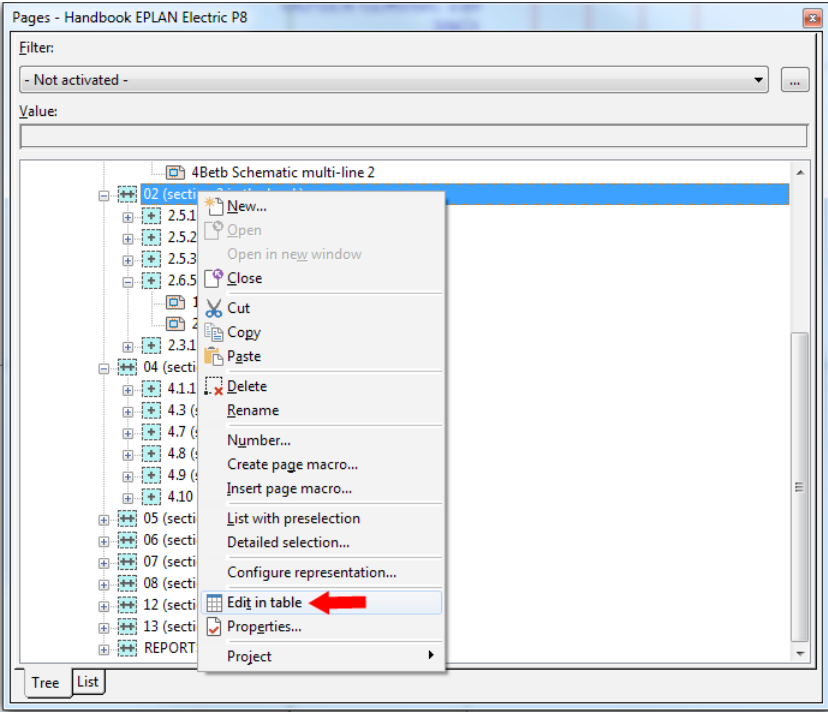


Fig. 4.56 Edit in table popup menu

Afterwards, EPLAN opens the **EDIT FUNCTION DATA** dialog. Aside from the actual edit function, this dialog also offers again the option to activate specific filters, thus editing quickly select properties of the objects contained on these pages.

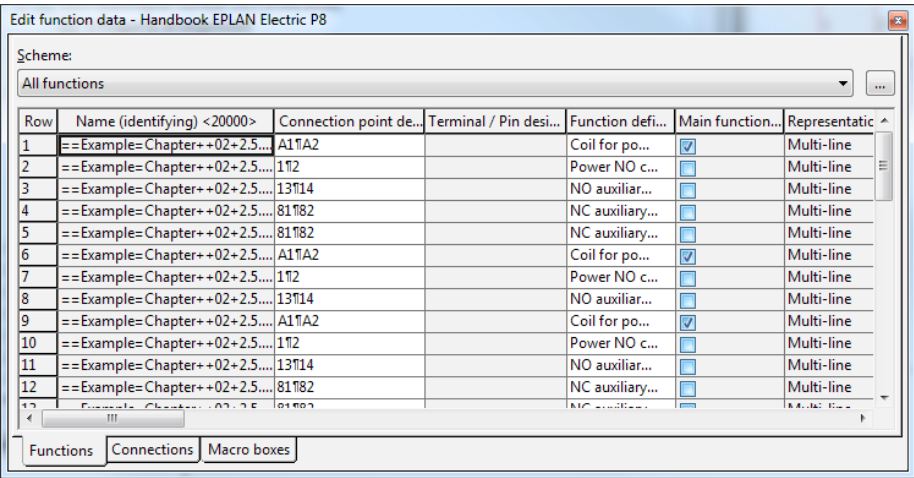


Fig. 4.57 Edit function data dialog in the page navigator

4.1.5 List with preselection

The LIST WITH PRESELECTION function in the page navigator makes it possible to filter specific areas quickly and without further filters.

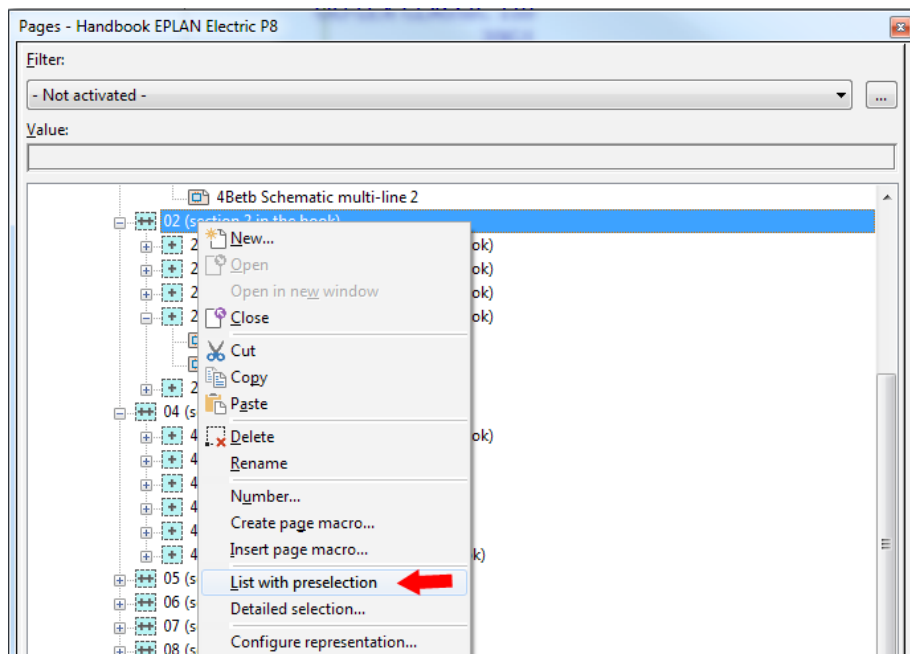


Fig. 4.58 Activating List with preselection function

For example, if you want to filter a specific identifier quickly, without creating an extra filter scheme, it is sufficient to select the identifier in the page navigator and right-click the LIST WITH PRESELECTION function.

EPLAN will then activate a “Selection” filter that is generated automatically. The page navigator will then only show the elements (pages in this case) that were selected before.

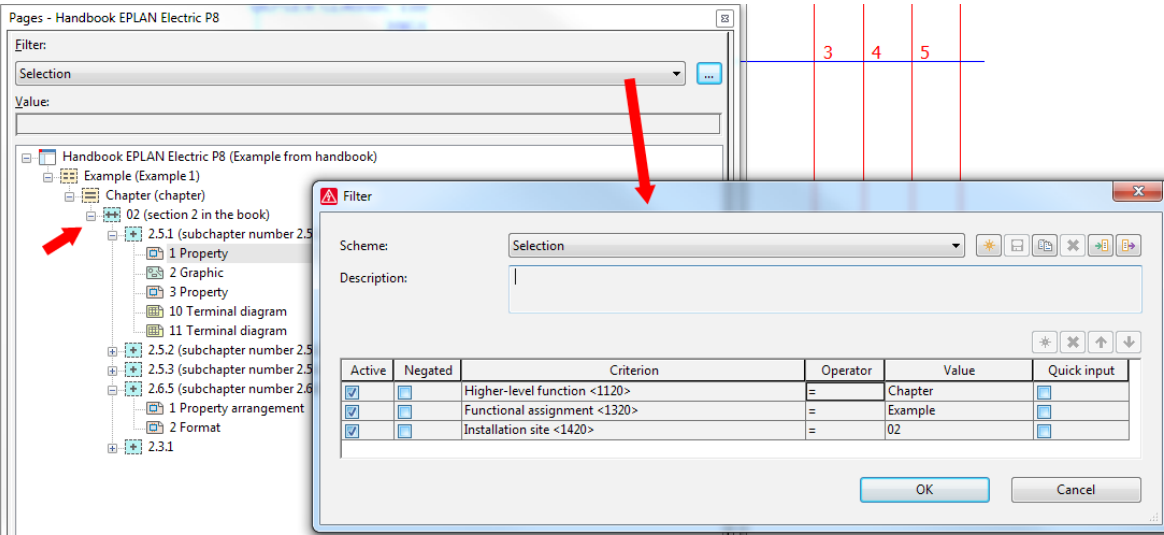


Fig. 4.59 Activated (automatic) “Selection” filter

4.2 General functions

The graphical editor can be divided into several areas. The main area is the working area for editing a schematic (the graphical editor). A number of dialogs can be associated with the main area. These range from the page navigator to user-defined toolbars.

The graphical editor is not restricted to a single representation. The familiar idea of **workspaces** can be used to rapidly switch the entire EPLAN user interface from one representation to another, including all the associated dialogs, toolbars, etc.

4.2.1 Title bar

Aside from the actual working area, there is the *Title bar*, which displays only little, but crucial, information. The most important function of the title bar is the display of the currently open project, which is displayed, by default, with the full path designation.

If so desired, a project property can be configured together with the project name. To do so, go to (OPTIONS/SETTINGS/USER/DISPLAY/USER INTERFACE menu) and modify the *Additional information on project name* item.

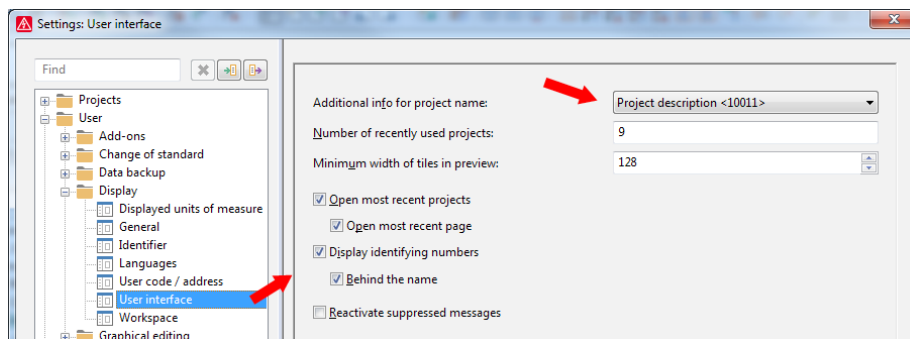


Fig. 4.60 Customizable display of the title bar

The following entries are possible here: user supplementary field 1, commissioning, project description, project number, supplementary field [1], or no project properties. In Fig. 4.61, the **None** setting has been selected.

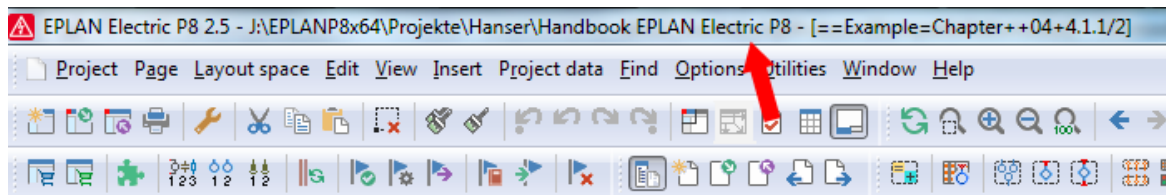


Fig. 4.61 Sample setting

You cannot change or add to the content of the title bar.

4.2.2 Status bar

The status bar in the graphical editor constantly shows various pieces of current information.

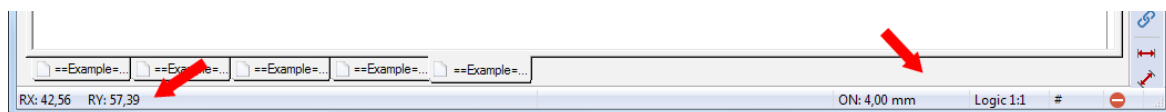


Fig. 4.62 Status bar

For example, the RX and RY coordinates of the current cursor position, information on the current grid (here 4.00 mm), and whether or not the grid is switched on (here ON), are all displayed here.

In the status bar, EPLAN also displays supporting information for the function currently in use (in this case, a graphical element is to be drawn). This does not apply to all functions, but looking at the status bar can be a great help for some EPLAN actions.

The last characters in the status bar have the following meaning:

#	The hash means that the project contains connections that have not been updated. ¹⁾
*	The asterisk means that the current page contains connections that have not been updated. ¹⁾
+	This character is displayed when the “advanced” support-log mode has been enabled (using the “ELogFileConfigTool.exe” tool supplied by EPLAN).

¹⁾ Since it is possible to assign menu commands to keyboard shortcuts in EPLAN, I would recommend assigning the **PROJECT DATA/CONNECTIONS/UPDATE** menu command to the **F11** key. This is easy to remember and the command then only requires a single keystroke.

4.3 Coordinate systems

The EPLAN coordinate system has the following structure. There are different points of view: the graphical system and the logical systems for the areas of electrical engineering, fluid power and process engineering. Which coordinate system is used depends on the respective page type.

4.3.1 Graphical coordinate system

The graphical system has its origin (X,Y = 0,0) at the lower left of the page. This means that when the page scale changes, the page is enlarged upwards and to the right.

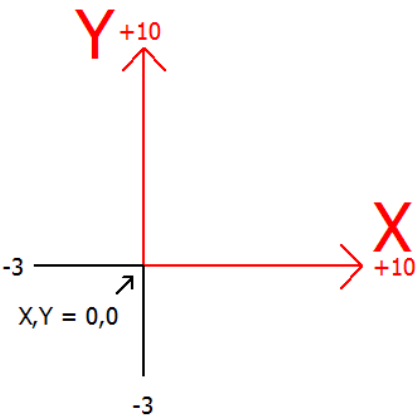


Fig. 4.63
Graphical

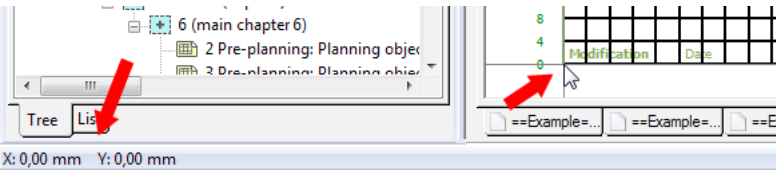


Fig. 4.64 Display in the status bar

4.3.2 Logical coordinate system

The logical system has its origin ($RX,RY = 0,0$) at the upper left. These positions are always measured from the original element and its origin (usually the so-called insertion point) in different directions.

In this case these are negative values, since the property is moved downwards and to the left based on the coordinate input X/Y in the property dialog.

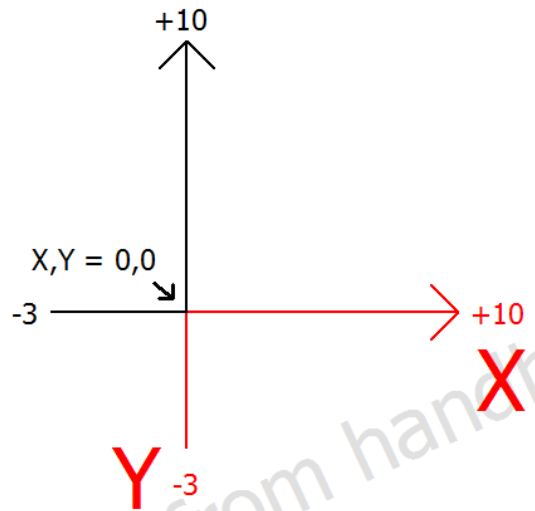


Fig. 4.65
Logical

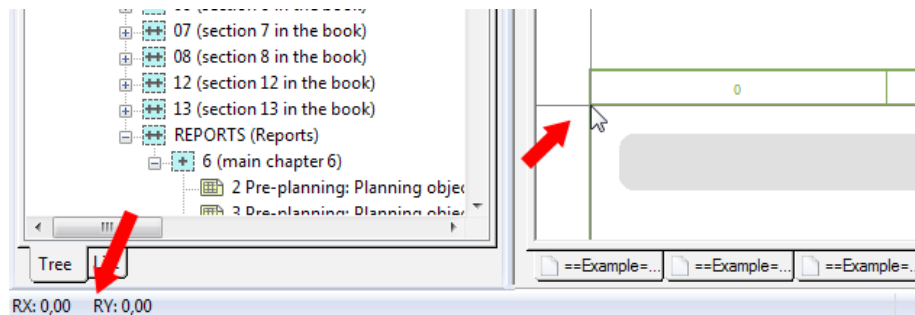


Fig. 4.66 Display in the status bar

4.3.3 Fluid power and process-engineering coordinate systems

Fluid power and process-engineering coordinate systems are two more examples that should be mentioned. For both systems, the origin is in the lower left ($RX,RY = 0.0$).

4.3.4 3D coordinate systems

Apart from the 2D coordinate systems mentioned so far, there is also a 3D coordinate systems in EPLAN (EPLAN Pro Panel add-on). It bears mentioning here for completeness' sake.

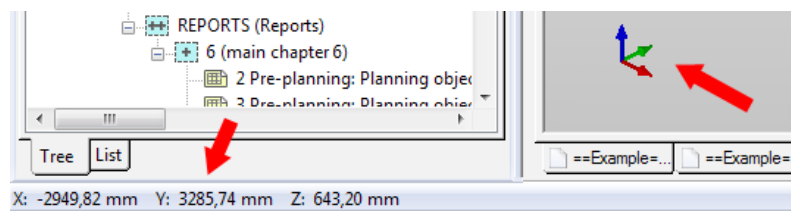
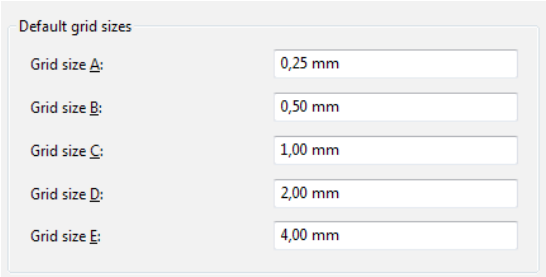


Fig. 4.67 Display in the status bar

4.4 Grid

In EPLAN, a fixed **grid** can be used to assist the user when placing (e.g. symbols) or when drawing (e.g. panel structures). The grid is visually displayed with small dots. The display of the grid can be enabled or disabled via the **VIEW/GRID** menu or the **CTRL + SHIFT + F6** short-cut key.

Fig. 4.68
Grid settings



Five different grids can be entered in the **OPTIONS/SETTINGS/USER/GRAPHICAL EDITING/2D** settings.

I recommend the following values for the grid settings: grid A = 0.25 mm; grid B = 0.5 mm; grid C = 1 mm; grid D = 2 mm and grid E = 4 mm.

These grid settings can then be selected in the **EDIT/OTHER/GRID [A to E]** menu or via the **VIEW** toolbar.

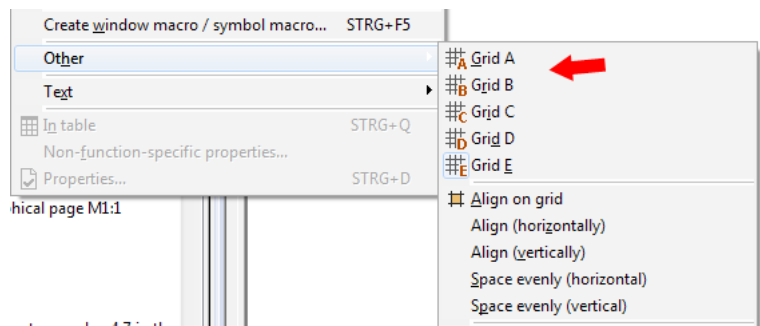


Fig. 4.69 Other menu item, with grid settings

Some of the grid settings should be assigned their own shortcut keys. This would allow for the page grid to be switched quickly, without having to switch to the grid all the time via the menu.

The **VIEW** toolbar has other buttons that will be briefly explained here. They can all be assigned to a keyboard shortcut as well.

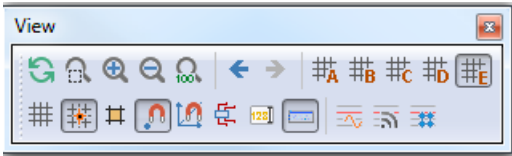


Fig. 4.70
Toolbar view

	This button turns the grid on or off.
	This function enables or disables Snap to grid. When Snap to grid is enabled, elements can only be placed on the activated grid. Intermediate positions or free placement are not possible. When the grid snap is switched off, then all elements can be freely placed (this also applies to logic elements such as symbols).
	This Snap to grid function allows you to align different elements (outside the grid) to the currently set grid. ¹⁾
	During drawing, the object snap will let you, for example, automatically link a line to insertion points (logical objects) or element points (graphical objects) (let them "capture" it).
	The design mode allows you to align graphical elements to specific points, i.e. to place them on specific coordinates. If the design mode is enabled, first you select the action, Move for example, and then you select the object and define its starting and end points.

¹⁾ The procedure is as follows: Click the button on the **View** toolbar. Then use the mouse to pull a window around the misaligned elements. The window can be pulled in any direction. When finished (all elements are contained in the window), release the mouse button and EPLAN will align all elements that are inside the window to the grid. It also works the other way around. First select the object(s) you want to align and then click the button. EPLAN will then align the selected objects to the set grid.

4.5 Increments, coordinate input

In addition to the grid settings, EPLAN also allows the input of increments and coordinates. These two features relate to the cursor position.

4.5.1 Increment

The increment is set via the **S** key or in the **OPTIONS/INCREMENT** menu. EPLAN opens the **SELECT INCREMENT** dialog.

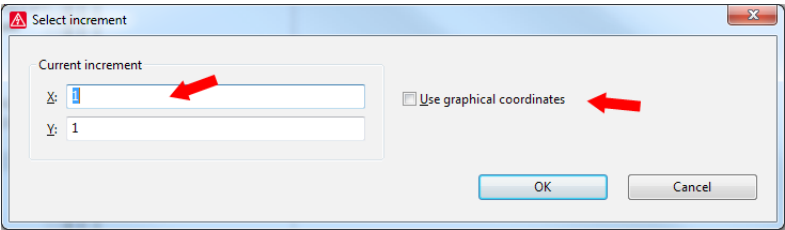


Fig. 4.71 Integer increment setting

The increments for the cursor can be entered into the **Current increment X and Y** fields of this dialog. If the **USE GRAPHICAL COORDINATES** option is deactivated, then it is only possible to enter integer values, such as 1, 3 or 17.

If the **USE GRAPHICAL COORDINATES** option is activated, then it is also possible to enter real numbers such as 1.75 or 3.88 because the graphical coordinates are then evaluated.

Increment input

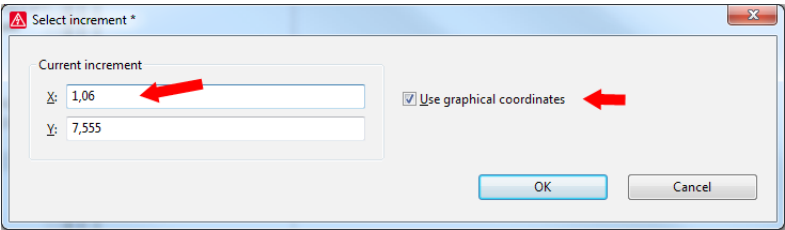


Fig. 4.72 Real-number increment setting

EPLAN also displays the information in the status bar, regardless of whether in the **SELECT INCREMENT** dialog the **USE GRAPHICAL COORDINATES** option has been activated or deactivated.

4.5.2 Coordinate input

Coordinates can also be directly entered into EPLAN to allow jumping to a particular point by clicking **OK**.

Coordinate input

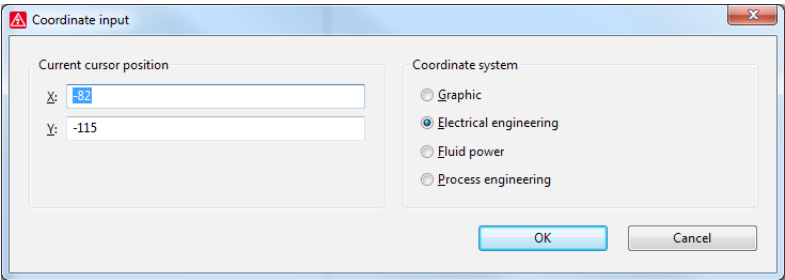


Fig. 4.73 Coordinate input

The **COORDINATE INPUT** dialog is started with the **P** key or via the **OPTIONS/COORDINATES** menu. The **X** and **Y** positions can be directly entered in this dialog. As soon as the dialog is closed using the **OK** button, EPLAN jumps to the specified **X/Y** coordinates.



NOTE: The mouse should not be used here because it would probably shift the position where you want to jump.

During coordinate input, it is also possible to switch between the different coordinate systems (graphic, electrical engineering, fluid power and process engineering).

4.5.3 Relative coordinate input

EPLAN allows both **static** (described in the previous chapters) and **relative** coordinate input. The **RELATIVE COORDINATE INPUT** function is accessed via the **OPTIONS** menu.

This function can be used to place the cursor (on which a line for placement might “hang”) in a new position. This means that you could start drawing a new line and then call the relative coordinate input. In the simplest case, you could enter the length of the line, and EPLAN would use this length, place the cursor in the position and terminate the drawing of the line. In contrast to the static coordinate input, the input here is always relative to the current cursor position.

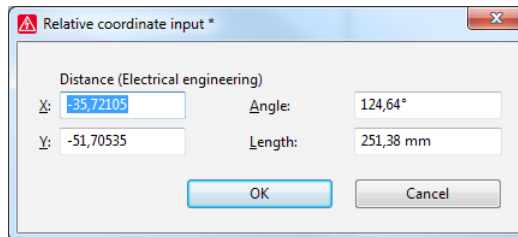


Fig. 4.74
Relative coordinate input



NOTE: If a grid is enabled and the jump to the cursor end point is not on the grid, EPLAN will still jump to the next grid point. You should take this into consideration when drawing.

4.5.4 Move base point

The **MOVE BASE POINT** function, accessible via the **OPTIONS** menu or the **O** key, sets a new base point (**X,Y = 0**), which can be manually set to be different from the **standard base point** (depending on the coordinate system, e.g. graphic, electrical engineering, lower or upper left).

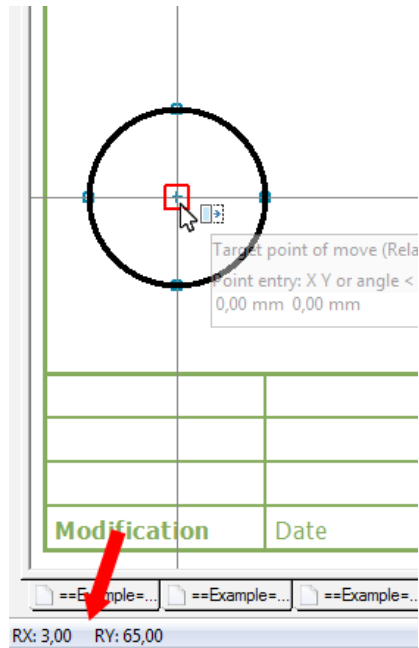


Fig. 4.75 Standard base point

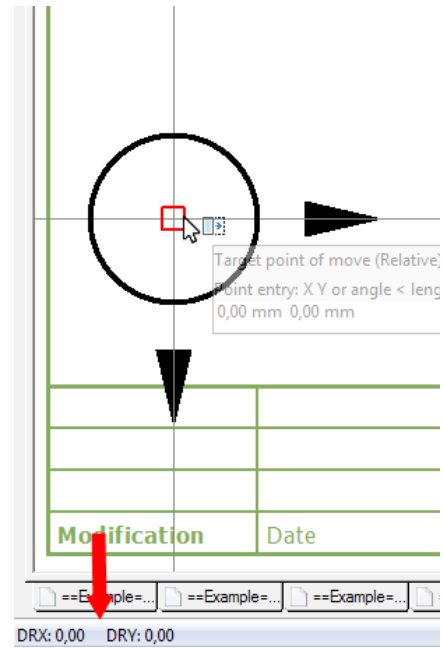


Fig. 4.76 New base point



In the example (center of the circle), the base point is $RX = 3.00$ mm and $RY = 65.00$ mm. In order to redefine the base point, for example, to calculate the horizontal values from the center of the circle, you place the cursor on the **center**, press the **O** key and then press **ENTER**.

EPLAN now generates a new current base point for this page with $X,Y = 0$. All positions are immediately calculated using this new base point and displayed in the status bar. To differentiate from the usual values displayed ($X = n$; $Y = n$), EPLAN displays the values of the base point shift with a starting D. In the status bar, the X and Y values are displayed as $DX = n$ and/or $DY = n$.

You can reset the base point shift using the **O** key or you can simply change pages.

■ 4.6 Graphical editing functions

In addition to schematics, EPLAN allows you to create a type of CAD drawing (CAD = Computer Aided Design). EPLAN also offers true to scale drawings with many dimensioning possibilities and many additional well-known CAD functions, such as stretching, trimming (modifying the length) and grouping of elements.

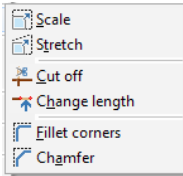


Fig. 4.77 Edit/
Graphic menu

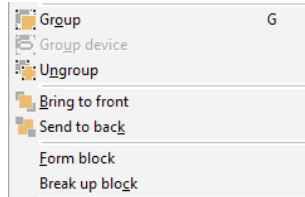


Fig. 4.78 Edit/Other menu

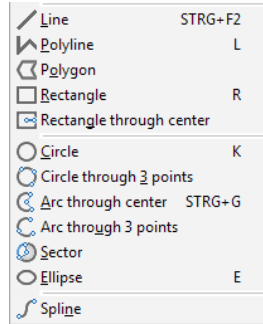


Fig. 4.79 Insert/Graphic
menu

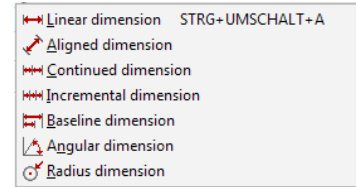


Fig. 4.80 Insert/Dimensioning
menu

These graphical functions can be accessed via the **INSERT/GRAPHIC [FUNCTION TYPE]** menu. Special functions such as stretching, grouping and converting a component to a graphic are located in the **EDIT/GRAPHIC** menu and others such as Group, Bring to front, etc. are located in the **EDIT/OTHER [FUNCTION TYPE]** menu.

The dimensioning functions are located in the **INSERT/DIMENSIONING** menu.

EPLAN remains a pure CAE system (CAE = Computer Aided Engineering), but the ability to work with CAD functions makes certain pure drawing tasks much easier.

4.6.1 Graphical objects: lines, circles, rectangles

With the *Circle*, *Rectangle*, *Line* or *Polyline* functions and many other options, EPLAN enables you to create graphical drawings. This will not be explained in detail here because the calls and the functions are very simple and easy to understand.



TIP: When using the graphical functions, it is a good idea to switch on the *Grid* or to enable the **SNAP TO GRID** function.

You should also “play” around with the page’s set grid. If you require 1 millimeter increments, then you should set the grid for this page to 1 (1 millimeter), either by using the buttons of the grid settings or directly in the page properties (i.e. via the **Grid <11051>** page property).



NOTE: After changing the grid, it is important not to forget to set the grid to the default grid again. Otherwise, it can happen quickly that schematic pages end up being created in the wrong grid, and then everything would have to be restored to the default grid.



To draw a rectangle, for example, you use the **INSERT/GRAPHIC/RECTANGLE** menu or the **R** key call up a rectangle and place the starting point on the drawing. Now there are three options.

Option 1: You draw rectangle using the mouse or the keyboard by counting grid increments. You can also adjust the height and width later in the rectangle's properties. EPLAN drawings are always accurately scaled. That is, 1 mm in reality is always 1 mm, regardless of the page's scale!

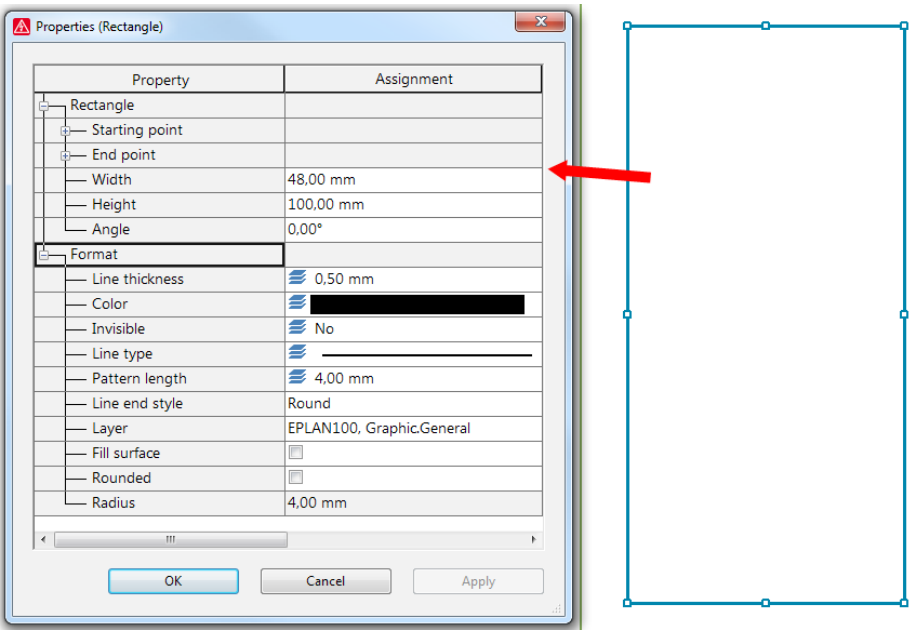


Fig. 4.81 Drawing a simple rectangle with manual width and height

Option 2: You start drawing the rectangle and define its size using the relative coordinate input (shortcut key **SHIFT + R**).

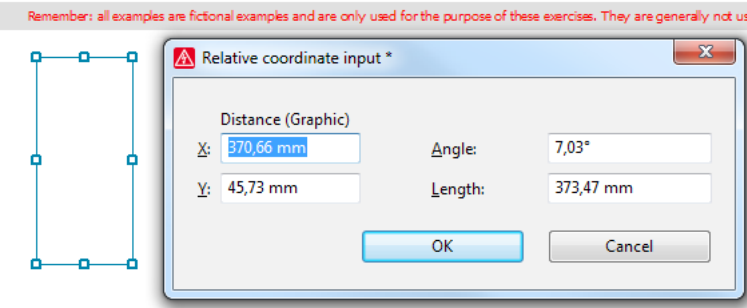


Fig. 4.82 Drawing a rectangle on the basis of relative coordinate input

Option 3: You use the **OPTIONS** menu to activate the **INPUT BOX** (always “hangs” on the cursor). Here, by directly entering values such as width and height, you can let EPLAN draw the rectangle.

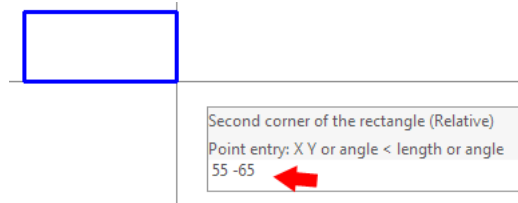


Fig. 4.83

Drawing a rectangle by direct input in the input box

4.6.2 Trim, chamfer, stretch and more

In addition to the pure graphical functions, such as drawing a rectangle or circle, EPLAN also lets you edit these graphical objects later. Such editing could include the following: changing the length of graphical elements, cutting out parts of graphical elements, rounding corners and much more. A few examples of the edit function in the **EDIT/GRAPHIC** menu will briefly illustrate the options.

Of course, some functions are not only intended for graphical editing, but also for ‘regular’ EPLAN work, such as the **ROTATE** function.

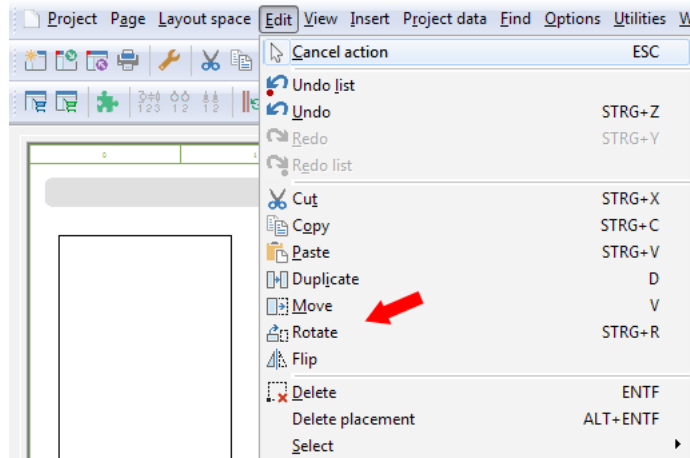


Fig. 4.84 Functions in the Edit menu



NOTE: The examples were partially carried out with the input box (in the **OPTIONS** menu) activated.

4.6.2.1 Rotate

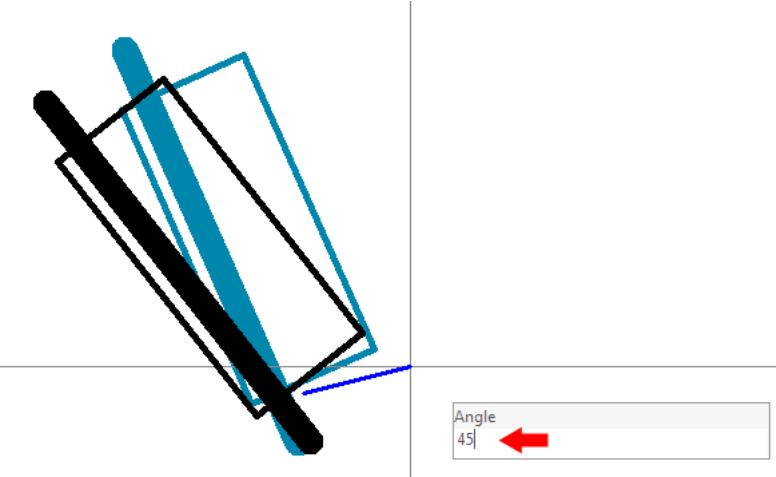


Fig. 4.85
Rotate

The ROTATE function allows you to rotate a selected object by any angle. It is called via the EDIT/GRAPHIC/ROTATE menu. Now, EPLAN waits for you to select the object to be rotated. Then, you have to define the center of rotation (around which the object is to be rotated). Now you enter the rotation angle directly in the input box and confirm with ENTER. EPLAN has rotated the object.



NOTE: Rotating is not limited to graphical objects. You can also rotate pure symbols (components). This eliminates the need to change the component variant in the properties (components) on the *Symbol/function data* tab. But you can only use the rotation angles that already exist as a symbol variant.

4.6.2.2 Mirror

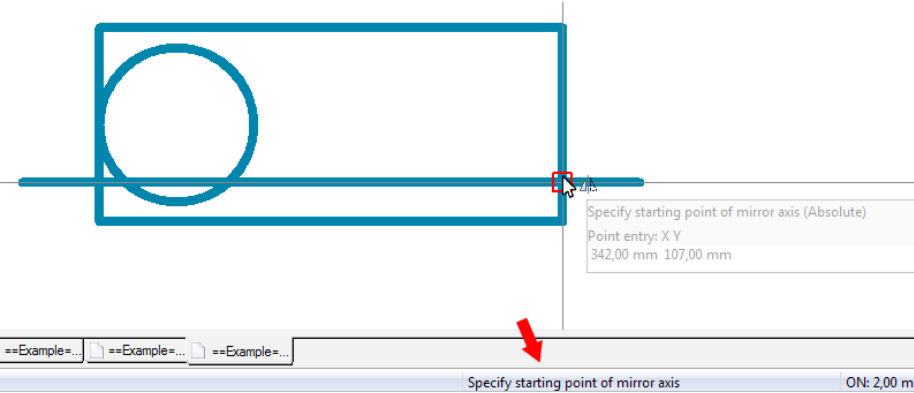


Fig. 4.86
Defining the starting
point of the mirror axis

Another function is the **MIRROR** function. This function is called via the **EDIT/MIRROR** menu. Then you select the object to be mirrored. Now EPLAN waits for you to input and/or define the *Starting point of the mirror axis*.

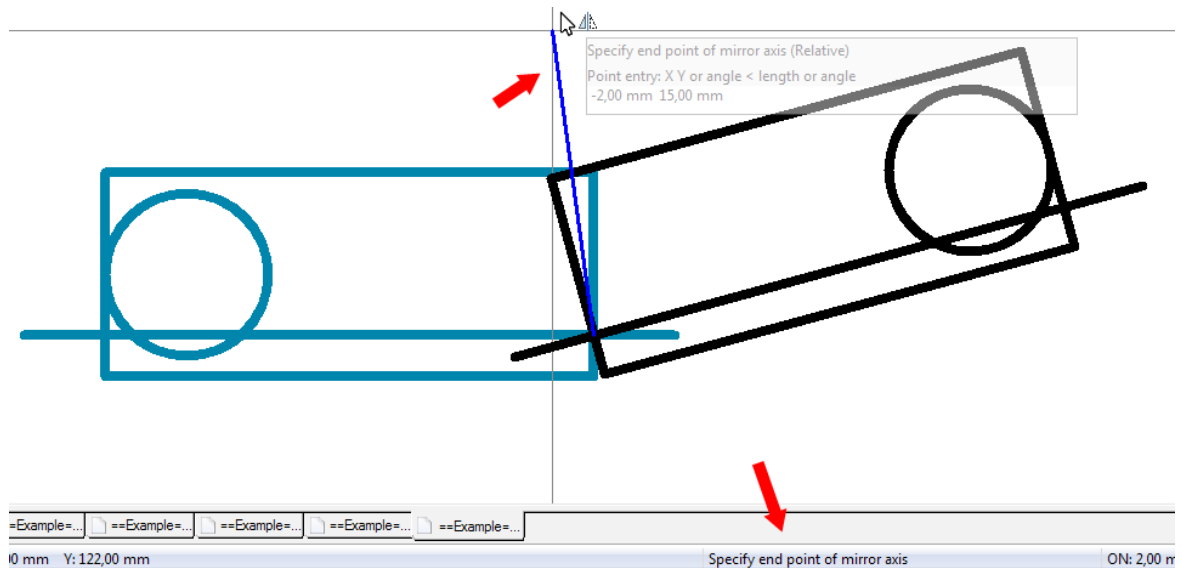


Fig. 4.87 Defining the end point of the mirror axis

After you have set the starting point of the mirror axis, you have to define the *End point of the mirror axis*. When you click the end point, EPLAN will mirror the object along the mirror axis defined.

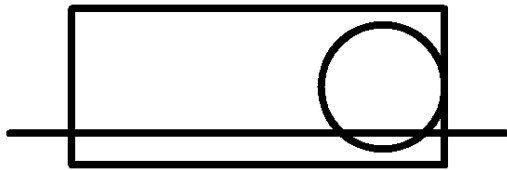


Fig. 4.88
Mirrored object



NOTE: Mirroring is not limited to graphical objects. You can also mirror pure symbols (components). This again eliminates the need to change the component variant in the properties (components) on the *Symbol/function data* tab. But you can only use the mirroring angles that already exist as a symbol variant.

4.6.2.3 Scale

Aside from rotating and mirroring, objects can also be enlarged or reduced. In EPLAN, these actions are summarized under the term **Scale**. The Scale function is called via the EDIT/GRAPHIC/SCALE menu.

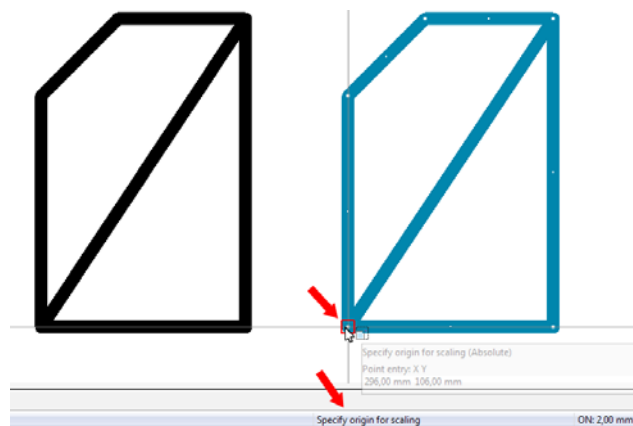


Fig. 4.89 Defining the origin for scaling

After you select the *Origin* you can click it. EPLAN opens the SET SCALE dialog.



Fig. 4.90 Setting the scaling factor

When you confirm this dialog with OK, EPLAN scales (here: reduces) the selected object.

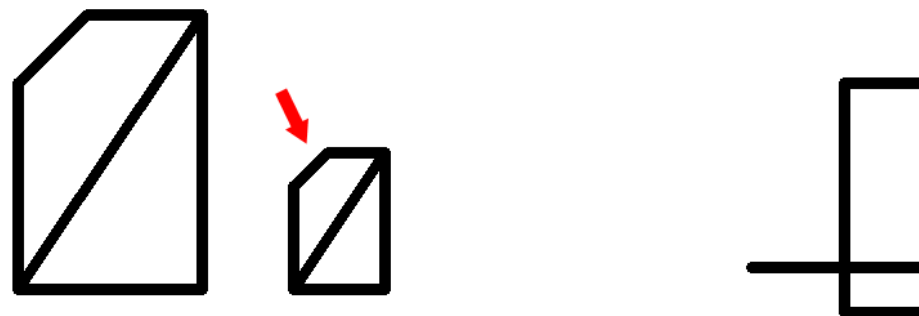


Fig. 4.91 Scaled object

When using the **Scaling factor** note that EPLAN expects a factor less than 1 to reduce (half the size smaller, for example, which is 0.5) and a factor greater than 1 to enlarge (for example, larger by a quarter, which is 1.25). You must enter a decimal point in the scaling factor for it to work.

4.6.2.4 Stretch

The **STRETCH** function allows you to lengthen and shorten objects. It is called via the **EDIT/GRAPHIC/STRETCH** menu. Then, you select the right-hand section of the rectangle (in this example) with a window. EPLAN places round markers in the corners and waits for you to define the starting point of the stretch.

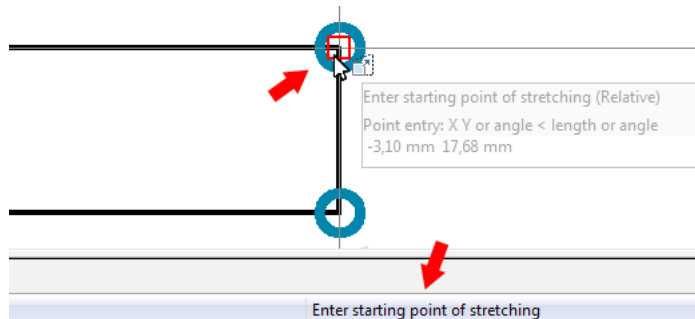


Fig. 4.92

Defining the starting point of the stretch

Once you have defined the **Starting point**, you use the mouse or keyboard to set the end point.

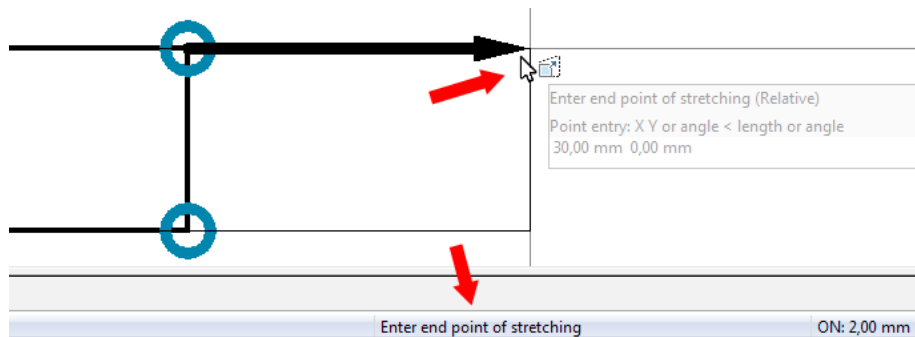


Fig. 4.93

Defining the end point of the stretch

After you have defined the **End point**, confirm it with **OK**, and EPLAN will execute the stretch.

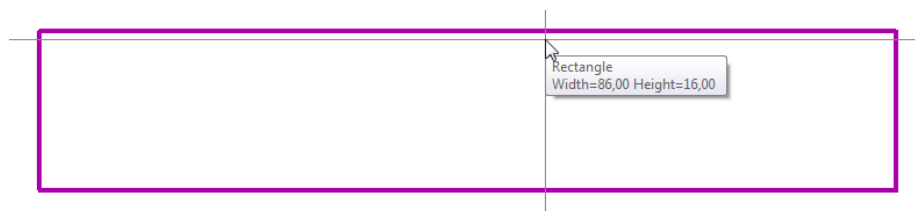


Fig. 4.94

Stretched object

Of course, rectangles cannot only be changed in length, but also in any direction.

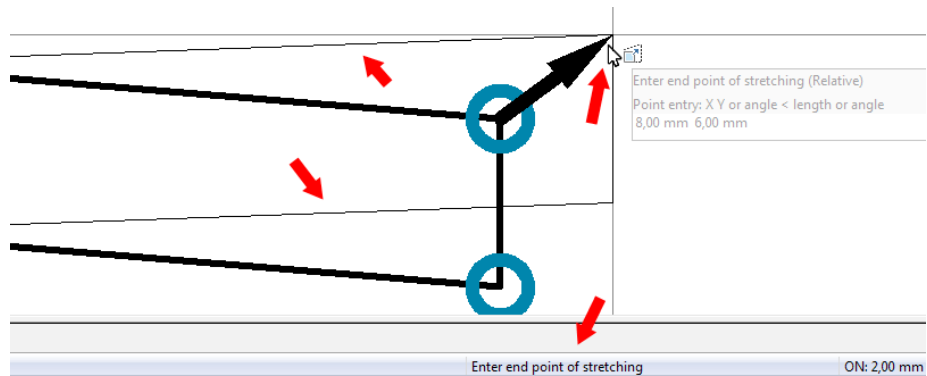


Fig. 4.95 Lengthening or shortening a different object (polylines)

4.6.2.5 Cut off

The **CUT OFF** function, also called **Trim** in the CAD world, is used to “cut” a so-called section out of elements. It is called via the **EDIT/GRAPHIC/CUT OFF** menu. After you have selected the function, EPLAN waits for you to select the section to be removed.

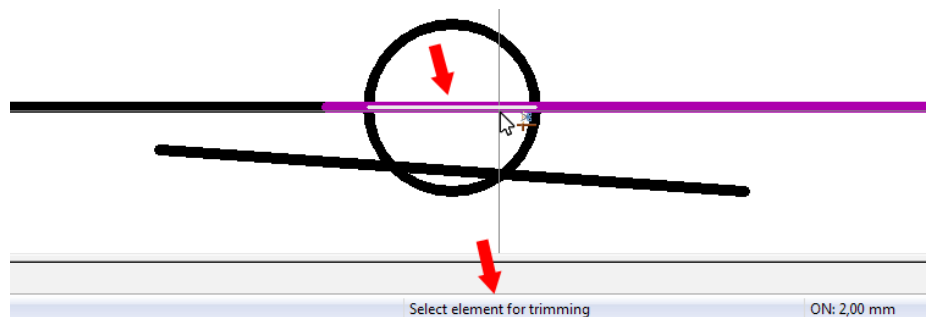


Fig. 4.96 Selecting the element to be removed

If, as shown in Fig. 4.96, you click the small section within the circle, EPLAN will cut it out. Then, the next section can be cut out, or the current **CUT OFF** function is terminated by pressing the **ESC** key.

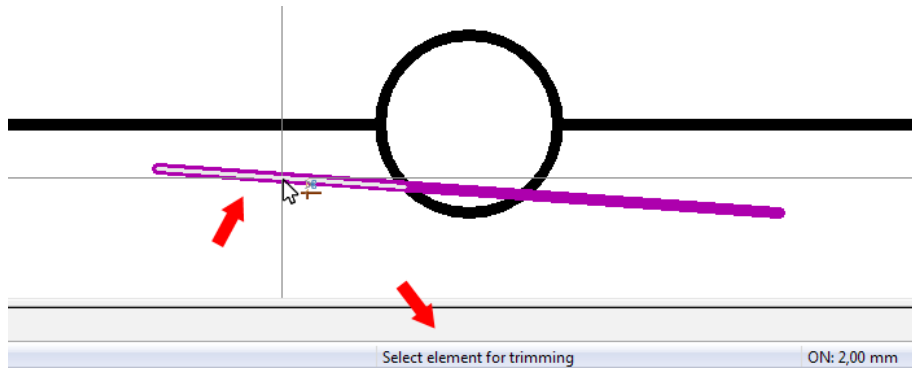


Fig. 4.97 Cut-off section

A typical example of this function would be the cutting off of sections of a populated horizontal rail or a mounting rail in a mounting panel layout.



NOTE: Using cutoff for grouped objects is not quite the same. If you want to cut sections off grouped objects, you have to keep the **SHIFT** key pressed and then cut the relevant section.

4.6.2.6 Change length

The **CHANGE LENGTH** menu item, as the name indicates, changes the length of objects. It is called via the **EDIT/GRAPHIC/CHANGE LENGTH** menu. Once this function has been called, you click the relevant object to change its length via the input box or by using the keyboard and/or mouse.

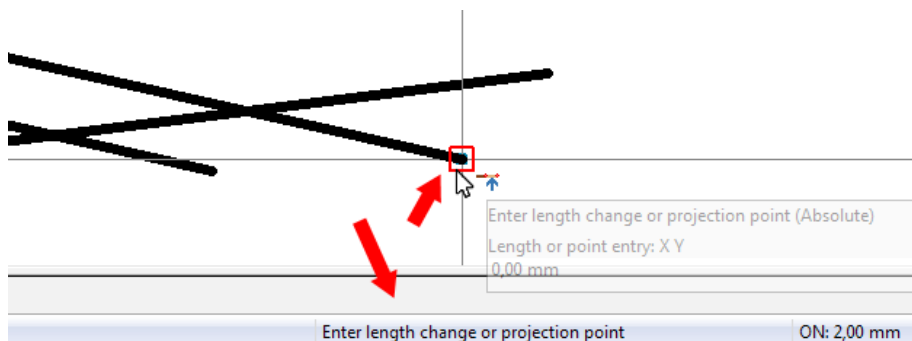


Fig. 4.98 Changing the length of a line

4.6.2.7 Fillet corners

The **FILLET CORNERS** function rounds off corners. It is called via the **EDIT/GRAPHIC/FILLET CORNERS** menu. After launching the function, you can click the desired corner and define the *fillet radius* in the input box.

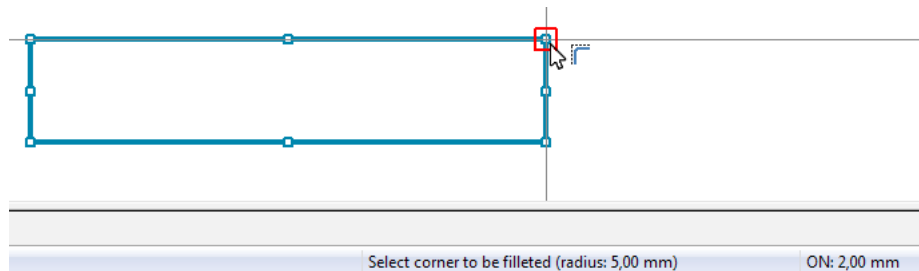


Fig. 4.99 Filleting corners

When you press the **ENTER** key, EPLAN fillets the corner according to the entered fillet radius.

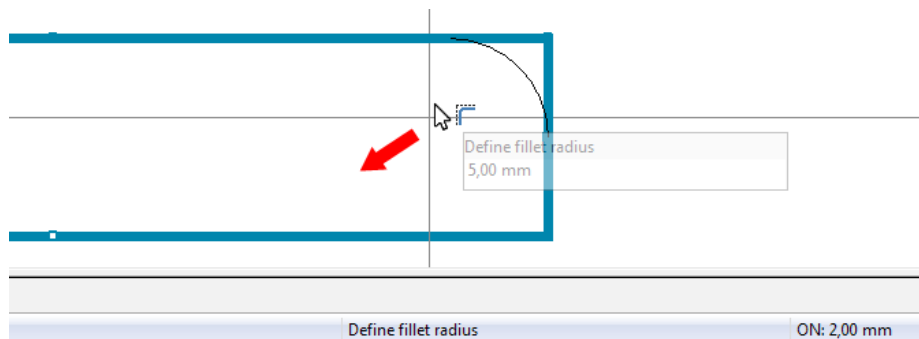


Fig. 4.100 Filleting the next corner

If you then click the next corner, it will be filleted automatically using the same radius.



EPLAN always remembers the radius most recently defined and applies it to subsequent actions.

4.6.2.8 Chamfer

The last function is **CHAMFER**. Chamfer does not fillet corners, it gives them a sloping edge. Similar to all other functions, it is called up via the **EDIT/GRAPHIC/CHAMFER** menu.

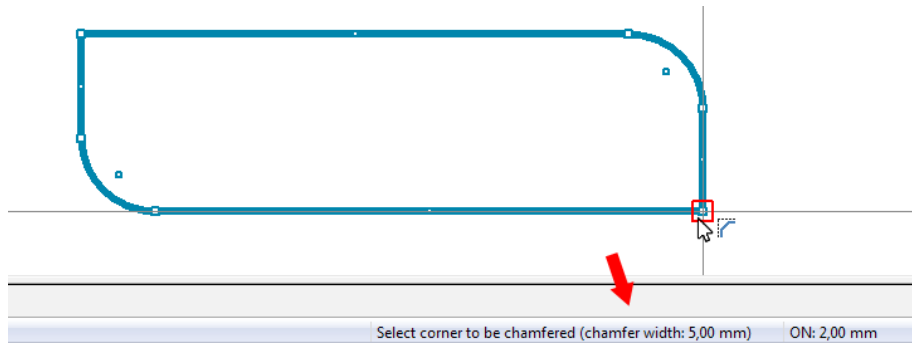


Fig. 4.101 Chamfering a corner

After starting the function, selecting the object and clicking the first corner, you enter your chamfer parameters directly in the input box. EPLAN uses these parameters after you press ENTER or left-click the object. The same chamfer setting can be applied immediately to the next corner.

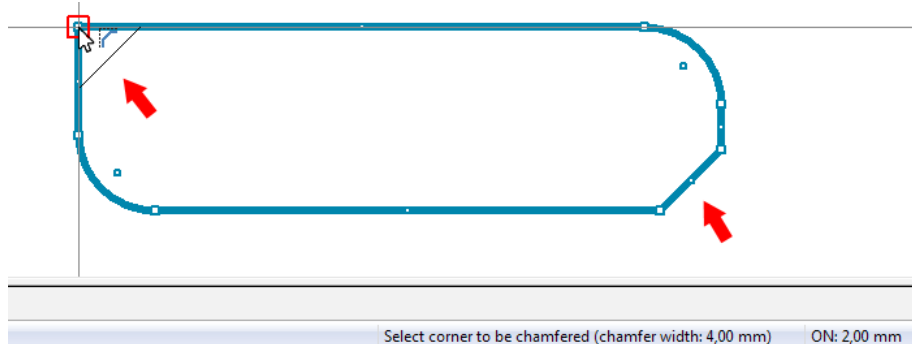


Fig. 4.102 Chamfer next corner

4.6.3 Group and ungroup

When you have several graphical elements, it can be difficult to move or, for example, to scale (enlarge, reduce) all these individual graphical elements. Every element must be manually adjusted to its new size via the **PROPERTIES** dialog, or when multiple elements are selected, the elements that are not to be scaled must be manually deselected.

With the **GROUP** and **UNGROUP** functions in the Edit/Other menu, EPLAN allows particular elements that belong together to be grouped and then be subsequently regarded as a single element. You use the mouse to select all elements that belong together and then call up the **GROUP** command from the menu (or alternatively use the **G** key).

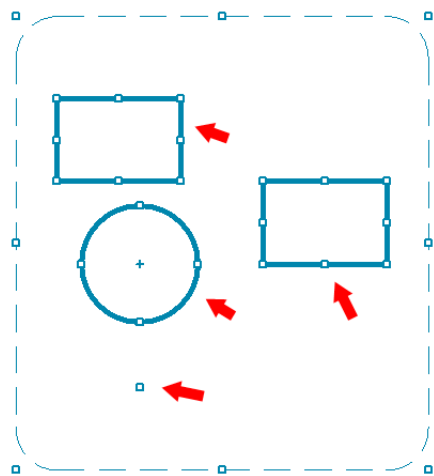


Fig. 4.103 Individual objects

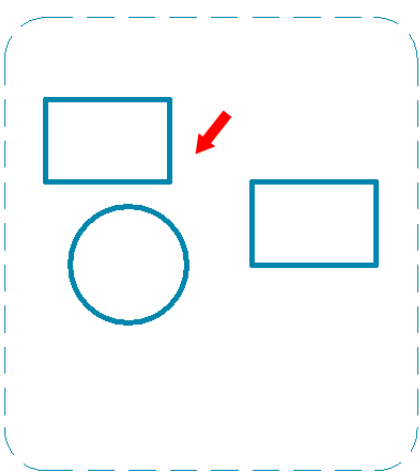


Fig. 4.104 Grouped objects

The grouping can, of course, be removed again. You simply select one of the grouped elements (EPLAN automatically selects all the elements in the group if you click anywhere on the grouped elements). In the **EDIT/OTHER** menu, you then select **UNGROUP**. EPLAN removes the grouping and all elements are once more individually accessible.

Individual elements within a grouping can also be edited independently. Press and hold the **SHIFT** key, and then double-click the desired object within the group. The object's property dialog opens, and the properties can be edited – without have to ungroup the objects. When all entries are complete, EPLAN applies the changes to this object. The remaining group objects are not affected.

4.6.4 Copy, move, delete

In addition to inserting and drawing new elements, you can also copy and delete elements. These functions are available in the **EDIT** menu. In this menu, you can call up the functions **COPY**, **MOVE**, etc. You can also use the keyboard to call up these functions: **CTRL + C**, for example, for **COPY**, or **CTRL + X** for the **CUT** function. To use the functions on elements, you have to first select an element (left mouse click) and then select the function.

	Cut	STRG+X
	Copy	STRG+C
	Paste	STRG+V
	Duplicate	D
	Move	V
	Rotate	STRG+R
	Flip	
	Delete	ENTF

Fig. 4.105
Editing functions

4.6.5 Dimensioning

In addition to actually drawing, you can dimension the drawn elements. EPLAN offers accurately scaled dimensioning, depending on the page scale that is set.

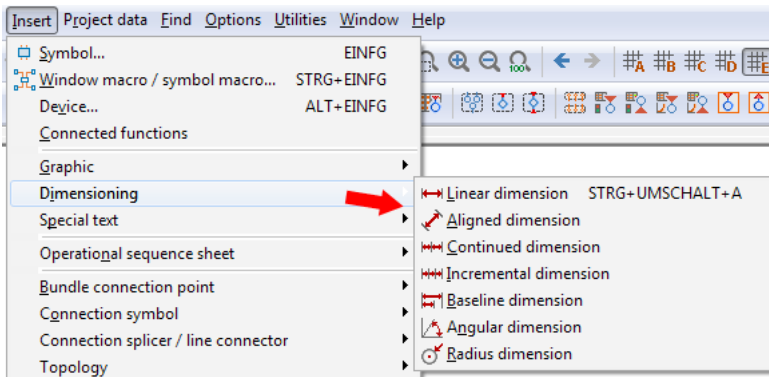


Fig. 4.106 Dimensioning functions

EPLAN comes with a number of dimensioning functions. Dimensioning types include, for example, continued dimension and radius dimension. The procedure for dimensioning elements is just as simple as drawing the elements themselves. The dimensioning types can be accessed in the **INSERT/DIMENSIONING [DIMENSIONING TYPE]** menu.

The various dimensioning methods will be illustrated with an example.



TIP: It is recommended that you enable the **OBJECT SNAP** function (accessible via the **OPTIONS/OBJECT SNAP** menu) when dimensioning. This allows EPLAN to start precisely at the ends (or the midpoints) of elements, without you having to tediously try to reach the ends of a line, for example. This usually cannot be done precisely.



In this example, a rectangle will be dimensioned. Use the **INSERT/DIMENSIONING** menu to call up the **LINEAR DIMENSION** function.

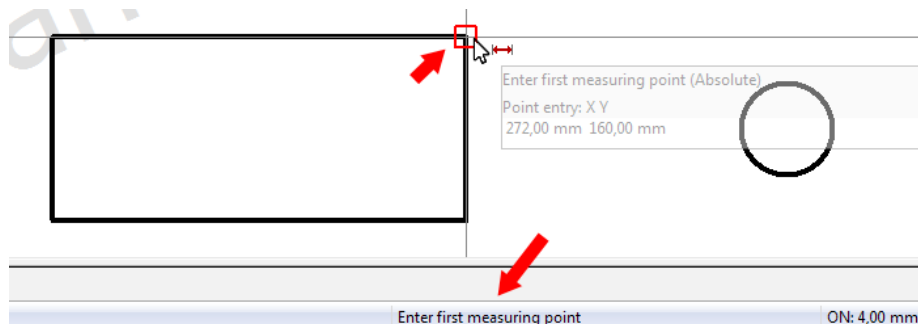


Fig. 4.107 Dimensioning starting point

Use the cursor to select the first **Dimension point** (corner of the rectangle) and then click it. EPLAN sets this point as the start of the dimensioning.

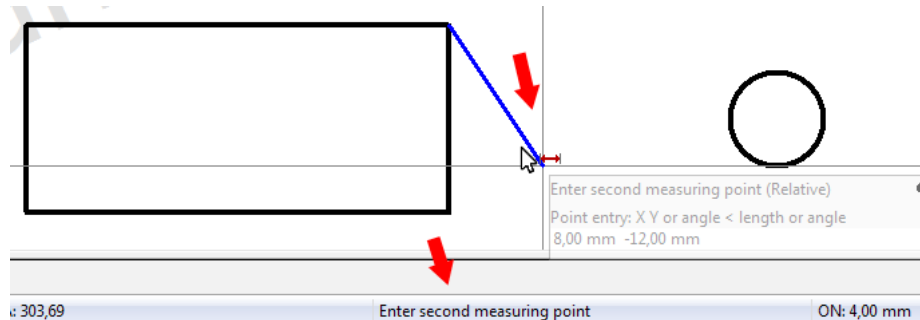


Fig. 4.108 Defining the second dimension point

Now look for the second dimension point. The first dimension “hangs” on the cursor and is placed using the mouse or with a click.

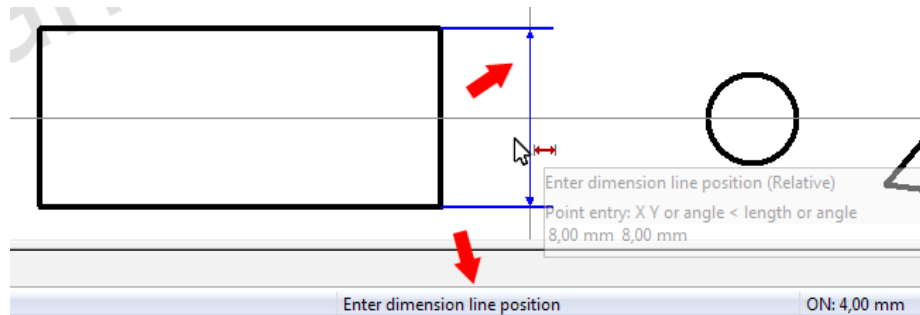


Fig. 4.109 Defining the position of the dimension line

Once the position of the dimension line has been set, EPLAN will dimension the object. For the dimensioning itself, there are a number of properties that can be called by double-clicking the dimension itself.

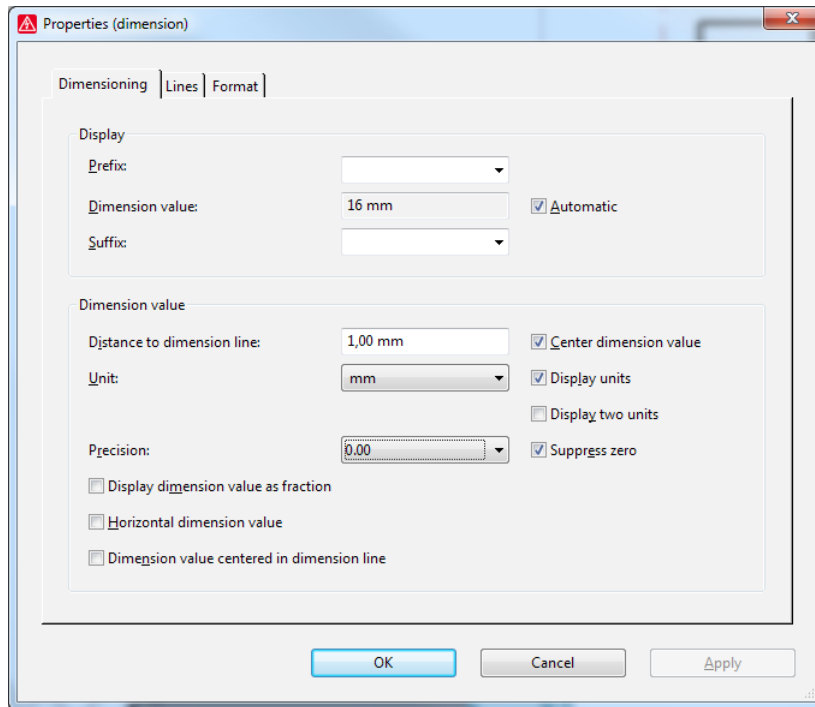


Fig. 4.110 Setting the current dimension

There are also additional tabs for *lines* and the *format* of the dimension labeling.

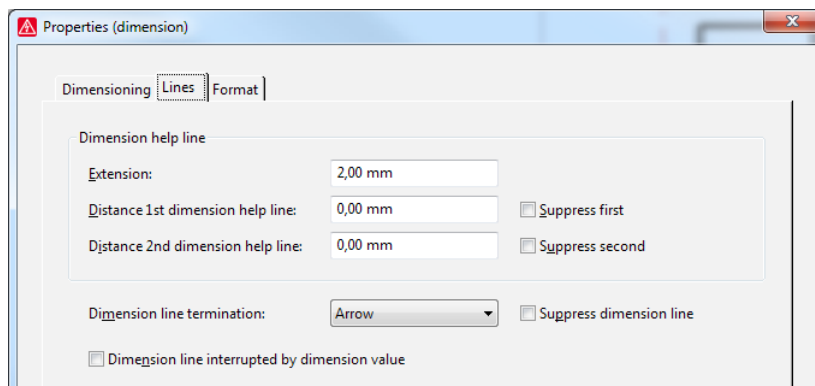


Fig. 4.111 Setting the dimension lines

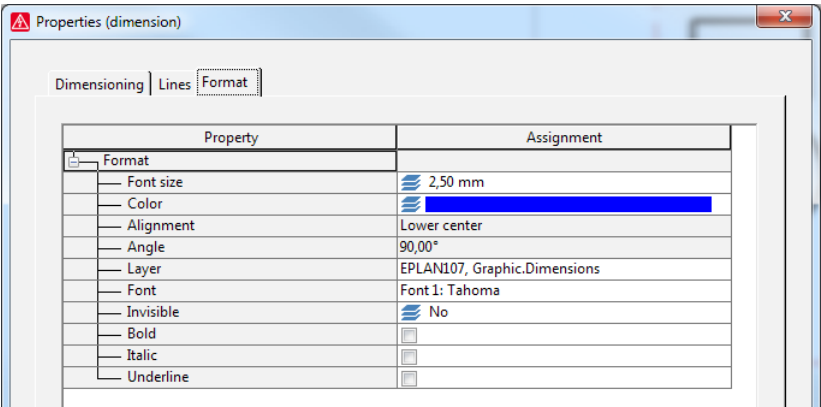


Fig. 4.112 Setting the format (dimension values, etc.)

The *Format* tab is where you define the settings for dimension values, etc. This does not affect the dimension lines. To control the dimension lines directly, you have to modify the **EPLAN107** layer in layer management.

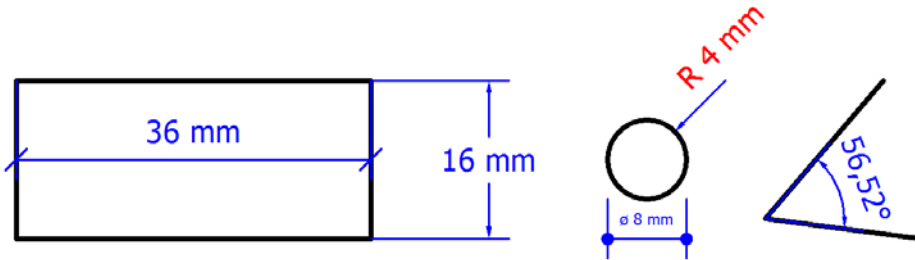


Fig. 4.113 Examples of different dimensions

4.7 Texts

Fig. 4.114
Free graphical text

In addition to graphical or electrical engineering functions such as symbols or devices, EPLAN also allows you to enter simple free texts (**INSERT/GRAPHIC/TEXT**).

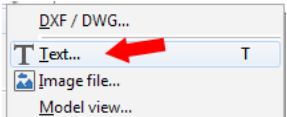
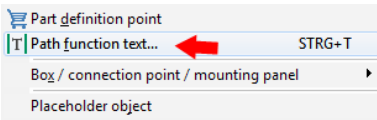


Fig. 4.115
Functional text in the
Insert menu

This is not a new feature but rather a necessity to allow various schematic functions to be clarified with text entries, and to provide a better explanation of the schematic and its functions, or to fill reports with additional information (or to have EPLAN fill them).



Plain text is always used in EPLAN. No numbering or coding system is used, so that the user can always see what the text is and does not need to look up a number in an encoded text file to know which text entry the number refers to.

4.7.1 Normal (free) texts

EPLAN allows the entry of normal (free) texts. Normal texts are texts that have no further functionality (except their visual display).

Translation of this type of text is possible without problems. Normal text should usually only be used to place text that does not need to be used in any other way, for example to provide extra information in a black box. Normal texts are inserted via the INSERT/GRAPHIC/TEXT menu or by pressing the T key.

It is only possible to later change normal (free) text to function text by activating the PATH FUNCTION TEXT selection box.

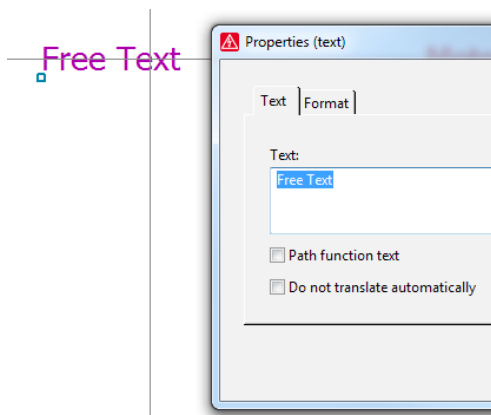


Fig. 4.116
Free text

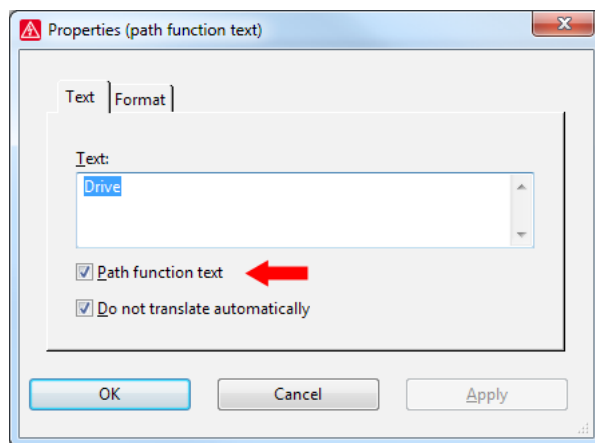


Fig. 4.117 Path function text selection box

4.7.2 Path function texts

In addition to normal (free) texts, EPLAN also has path function texts. These are equipped with “intelligence”. Path function texts should generally be used to automatically fill specific devices with function text, for example based on the page path.

Automatic path function text

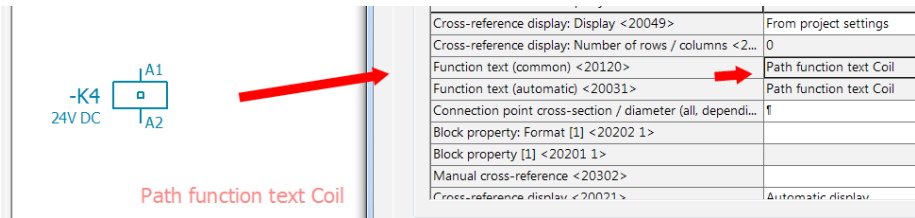
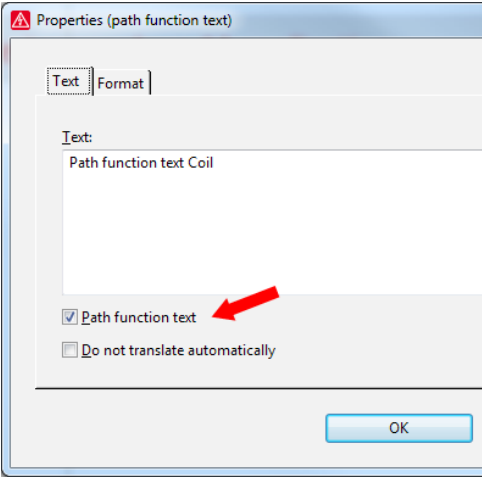


Fig. 4.118 Path function text

EPLAN uses the path function text for embedding in different reports (depending on the structure of the form) or for generating further reports. It is therefore advantageous to always work with path function texts, since subsequent conversion from normal texts to path function texts is possible, but requires unnecessary extra work.

Path function text is inserted via the **INSERT/PATH FUNCTION TEXT** menu. If path function texts are generally to be used, the keyboard shortcut for free text should be redefined to path function text. In this case, the keyboard shortcut for normal (free) text should be set to a different keyboard shortcut.

Fig. 4.119 Path function text



TIP: Normal (free) texts can be converted to path function texts by simply activating the *Path function text* setting in the **PROPERTIES – TEXT** dialog. After you exit and save the **PROPERTIES – TEXT** dialog, the layer is also automatically set to the standard layer for path function texts: **LAYER110. Graphic.Path function texts.**

To generate a report for a path function text for a device, it normally has to be directly (in the path) below or above the devices (the insertion points of the path function text and device are then “in sequence”).

But this might not be desired, for reasons of space or because the path function text is to apply equally to several devices. To avoid having to align the path function text several times in the path of the individual devices, you can set the **GRAPHICAL EDITING/GENERAL** project setting and, in this case, *Extend path function text on the schematic path* (always to the left and right).

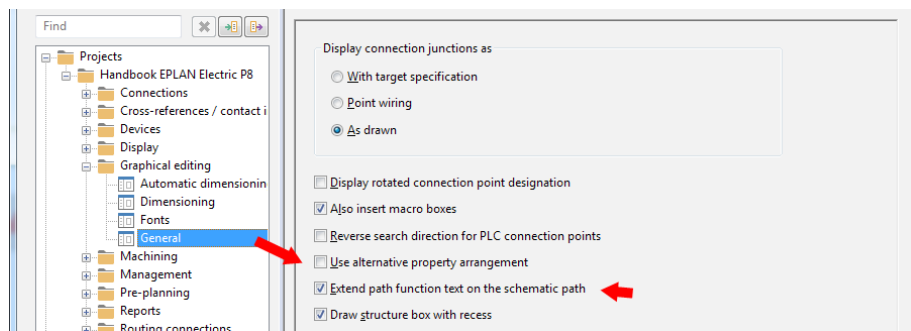


Fig. 4.120 Extend path function text setting

This way, it will not matter whether the insertion point is directly below the device or not. It is sufficient if the path function text and the device are located in the same path (the column).



NOTE: But extending the reporting of text to the path can also result in the text being assigned to objects in the path where this is not wanted. Therefore, you need to be careful when using the *Extend path function text on the schematic path*.

Apart from this extension setting, there is a second setting for path function texts that, however, concern only the PLC connection points. This setting is a project setting (*Device/PLC node*) and is deactivated or activated there.

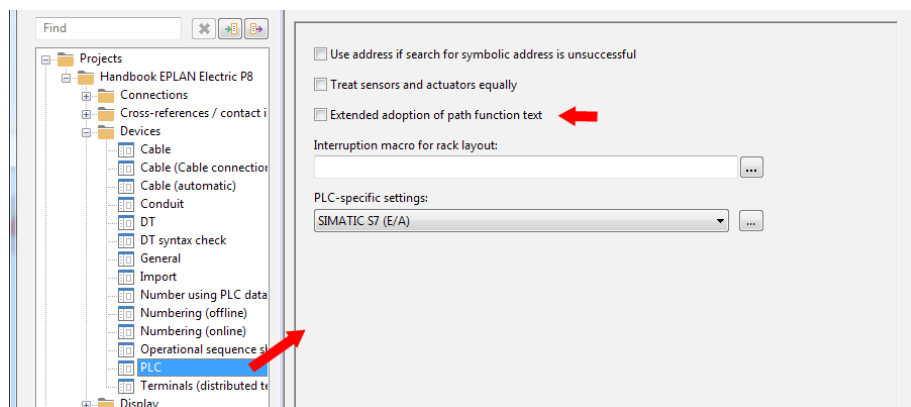


Fig. 4.121 PLC-extended function text import project setting

This is how it works when this setting has been activated: If EPLAN does not find a function text in the path, and no function texts (automatic) of sensors/actuators connected either, EPLAN will search the previous paths (search direction to the left) for function texts and import the first one found.

This setting should be checked should EPLAN transfer 'wrong' function texts to PLC connection points. It may be that a function text is missing in fact, or the existing text is not a function text.

4.7.3 Special texts

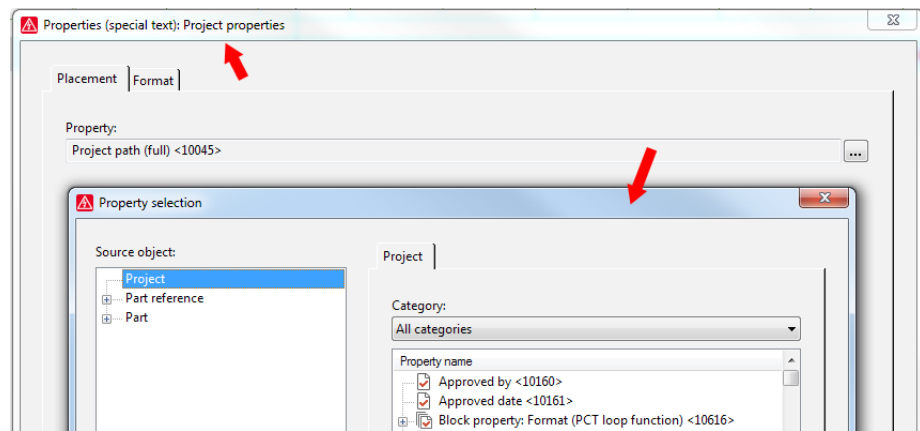


Fig. 4.122 Special texts

Special texts are another type of text used in EPLAN. Special texts can be *page property texts* or *project property texts*. They are generally used in forms (project properties) or in the plot frame (page properties) and less so as text on project pages.

Page property or project property texts are inserted via the **INSERT/SPECIAL TEXT/PROJECT PROPERTY** menu or **PAGE PROPERTY**. This way you can place text from the project properties or the page properties on pages.

The following example clarifies this application. In the project properties, 400 V AC for the main current voltage has been entered in **Supplementary field [10] <10901 10>**.

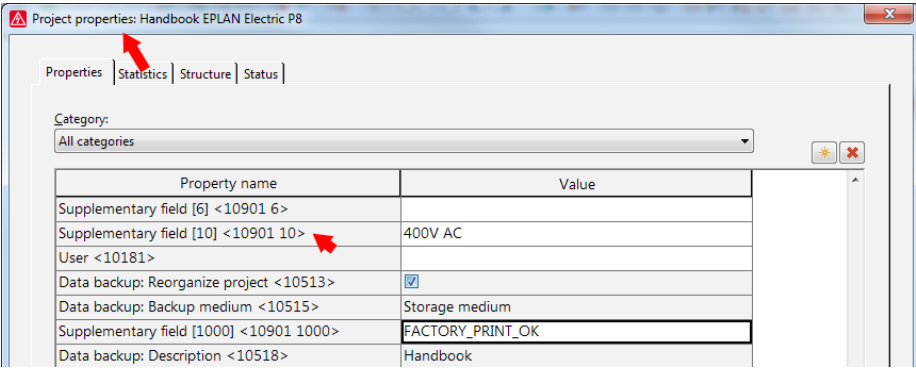


Fig. 4.123 Supplementary field [10] <10901 10> project property

This project property can now be used in the schematic, for example to show the main current voltage. The project property is called and placed via **SPECIAL TEXT/SPECIAL TEXT/PROJECT PROPERTY**.

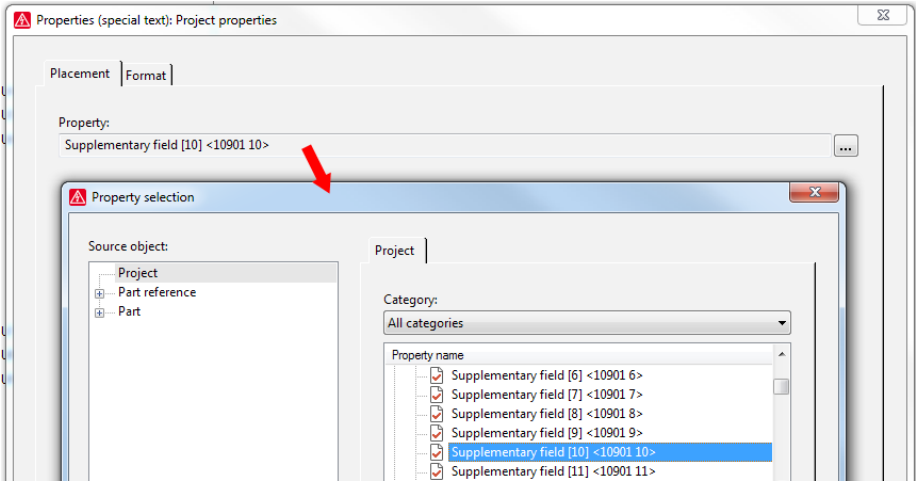


Fig. 4.124 Selecting project properties

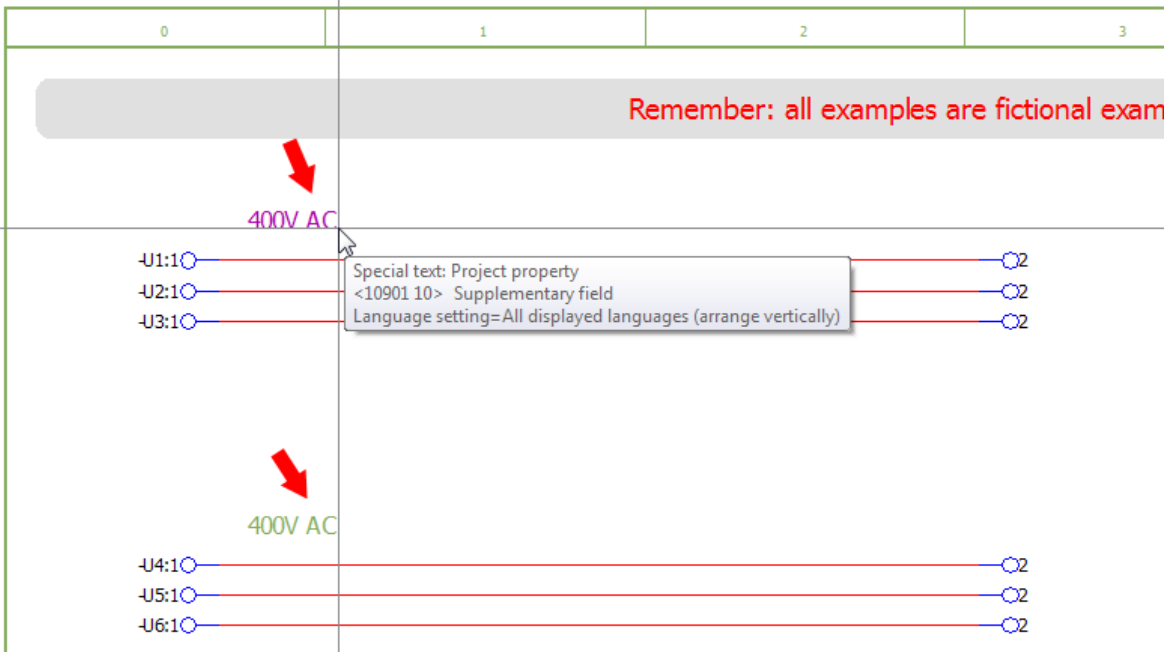


Fig. 4.125 Placed project property

If the voltage later changes or additional information is needed, the input in the project properties can simply be changed, and everywhere in the project where this project property is displayed will be automatically updated. This can save a lot of time in manually changing this information.

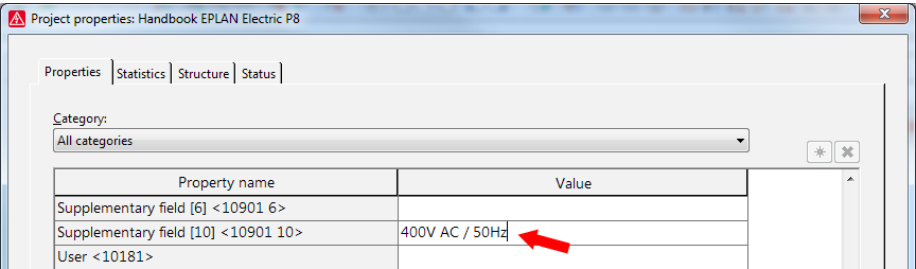


Fig. 4.126 Changed input in the project property